HUAHUA 桦桦数控®

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加工中心使用说明书

Manual of Machining Center



一站式定制家具方案

精准开料 | 高速裁板 | 自动封边 | 数控钻铣

CNC Router Machine | Computer Beam Saw | Edge Banding Machine | CNC Drilling Machine

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1 Attached files with the machine

During machine use, ensure that all accompanying documents are delivered to the customer in their entirety. These documents must be reviewed by any operator during equipment transportation, installation, commissioning, and maintenance. Store these documents in a suitable location for easy access by operators and maintenance personnel.

The machine comes with the following documents:

1、 User manual: It is divided into several chapters according to the model type, equipment safety, usage information and software operation details, based on the complexity of operation.

2、 Equipment gas path diagram.

2 Security Information

2.1 Operating Procedures

1. Before operating the machine, operators must be familiar with its performance , purpose , and operating precautions . Personnel who have not received systematic training are strictly prohibited from operating the machine alone .

2. It is essential to check whether the grounding wire is intact and reliable .

3. Sufficient lighting should be provided for nighttime operations .

4. It is strictly forbidden to operate the machine while wearing gloves , necklaces, ties, or similar items .

5. It is strictly forbidden to operate the machine while sick , under the influence of alcohol, or while taking medication (as these conditions may affect reaction speed) .

6. This machine must be used by designated personnel only. Unauthorized use is prohibited.

7. Put on your work clothes , goggles , and mask , and check that your clothing is securely fitted with the "three tight" measures: tight collar, tight clothing, and tight fit.

The hem of the top is tight.

8. Check the work area for debris to prevent slipping and tripping .

9. Before starting the machine, check whether the safety guard is in a reliable position and check whether there are any tools or foreign objects placed on each part.

10. Before operation , check whether the cutting tool is deformed or cracked , whether the cutting edge is clean , and whether the cutting tool is fully locked .

1.1 . When the tool is running , do not set the tool speed to exceed the maximum speed .

1. After machine maintenance, all safety devices must be installed before restarting.

13. If a malfunction occurs during machine operation, the machine must be stopped immediately. After the machine stops, the power should be turned off and warning signs should be set up. Non-professional maintenance personnel must not disassemble the equipment without authorization.

14. If an operator needs to leave their post , they must first turn off the machine .

1.5 . During machine operation , do not approach or touch rotating or moving parts .

1.6 . Dust and debris must be cleaned regularly.

1.7 . When the work is finished, turn off the machine switch and disconnect the power. Wait at least 2 minutes after the machine has completely stopped before approaching the electrical equipment.

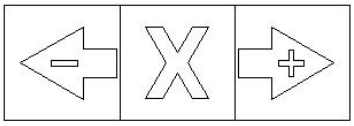
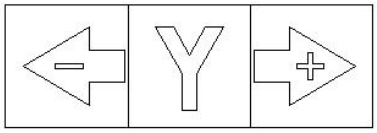
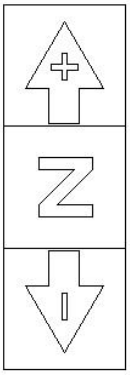
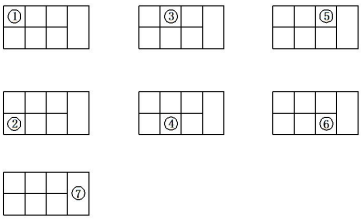
2.2 Symbols

Depending on the machine tool structure , the following lists the hazard and instruction markings that may appear on the machine .

2.2.1 Warning signs

logo	Label Description
	Beware of high-speed cutting tools !
	Be careful when pressing down !
	Caution: Collision warning !
	Be careful when pressing down !
	Beware of electric shock.
	Danger

2.2.2 Instruction Identifier

logo	Label Description
	X-axis direction label
	Y-axis direction indicator
	Z-axis direction label
	Valve label

3 Technical Information

This chapter mainly describes the machine type , main components , equipment parameters, safety devices, and working devices .

Machine Description :

The 912 series panel saw is a machine used for processing wood panels . It has functions such as panel cutting, drilling, milling, etc. It is a machine tool equipped with a program control system. Do not use it for any purpose other than its original settings.

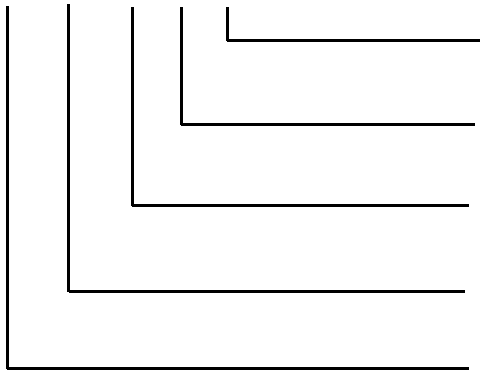
The following materials are prohibited from being processed:

1. Gypsum board
2. Resin glass, polycarbonate foam
3. Glass
4. Metal
5. Plastic sheets. (This list does not include all materials; please consult the manufacturer for specific processing permissions.)

1. All illegal applications are prohibited. Unauthorized applications can affect the stability of the equipment and may even cause malfunctions and damage, posing a risk to the health and safety of operators.
2. The manufacturer/supplier shall not be liable for any losses caused by illegal applications.

3.1 Model Description

9 12 M Z Y



Y : Pressure roller unit (optional)

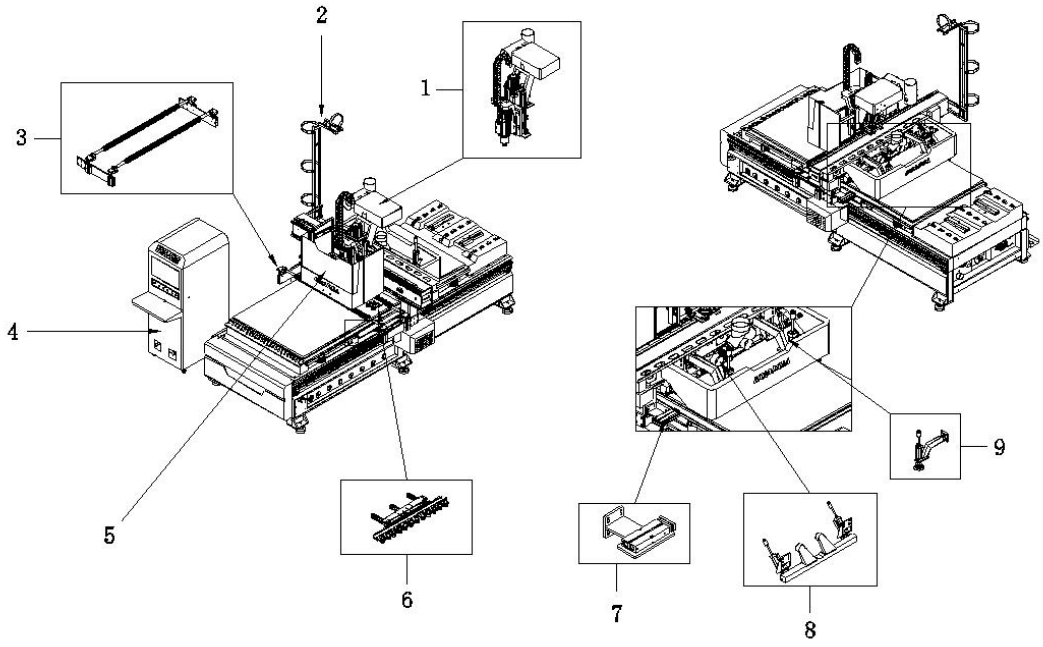
Z : Automatic feeding unit (optional)

M: Regular style

12 : 12 knives

9 : 9 series panel saw

3.2 Main components

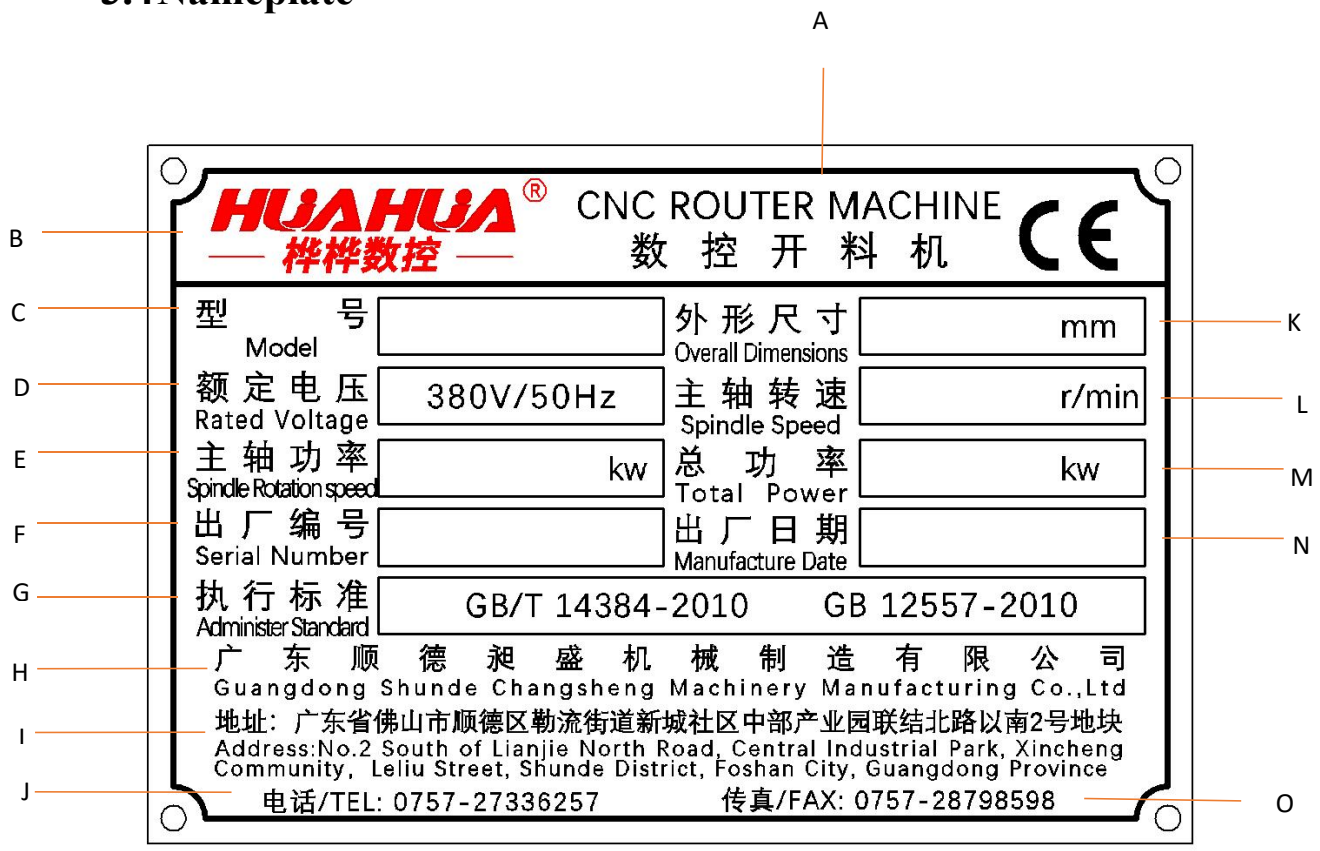


Serial Number	name	Serial Number	name
1	Cutting components	6	Straight-line tool magazine assembly
2	Vacuum pipe support assembly	7	Side-push assembly (optional)
3	Pressure roller assembly (optional)	8	Material pushing and dust collection components
4	Electrical box assembly	9	Automatic feeding component (optional)
5	Shield components		

3.3 Technical parameters

Serial Number	project	912MZY
1	spindle speed	18000 r/min
2	Spindle power	9 kW
3	External dimensions	3950*3324.1*2905
4	Rated voltage	380VAC
5	Servo power	4KVA
6	Workbench height from the ground	910 mm
7	Compressed air pressure	0.6～0.8 MPa
8	Vacuum negative pressure	-0.4～-0.9mpa
9	Maximum machining size	2 800 * 1 300
10	vacuum region	7
11	Idle speed	X: 6 0m/min Y: 6 0m/min Z: 2 0m/min
12	Repeatability	0.05mm
13	Vacuum pump power	7.5 kW
14	Vacuum pump exhaust speed	250 m^3/h
15	Total power	2 0.5kW

3.4Nameplate



A. Equipment Name

B. Company logo

C. Equipment Model

D. Rated voltage

E. Spindle power

F. Serial number

G. Administer standard

H. Company Name

I. Company Address

J. Company contact number

K.Overall size

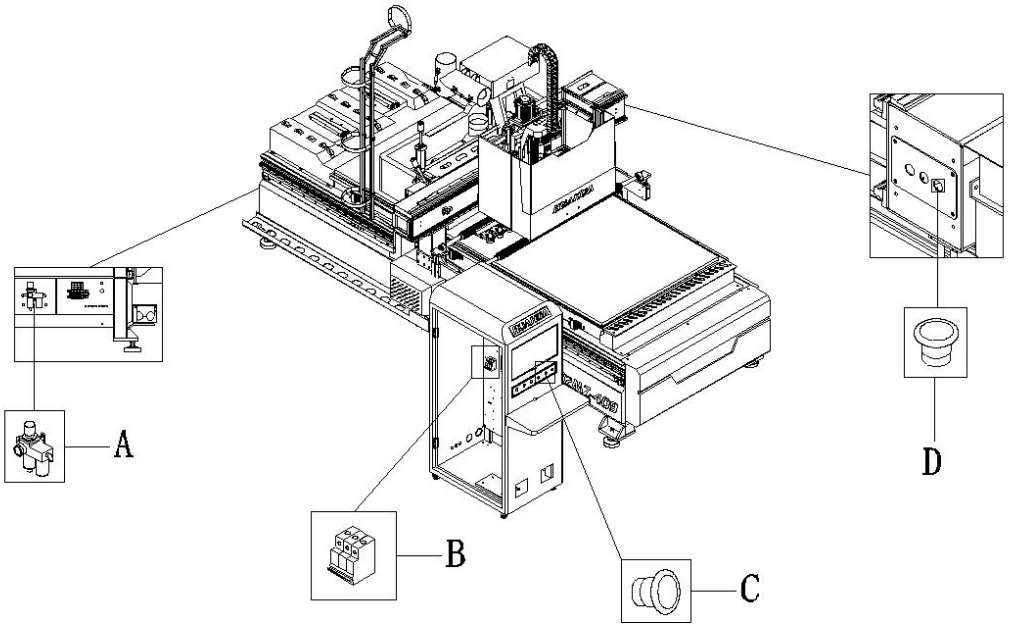
L.Spindle speed

M.Total power

N.Manufacture date

O.Company fax

4 Safety devices



Illustrations and descriptions of major safety devices

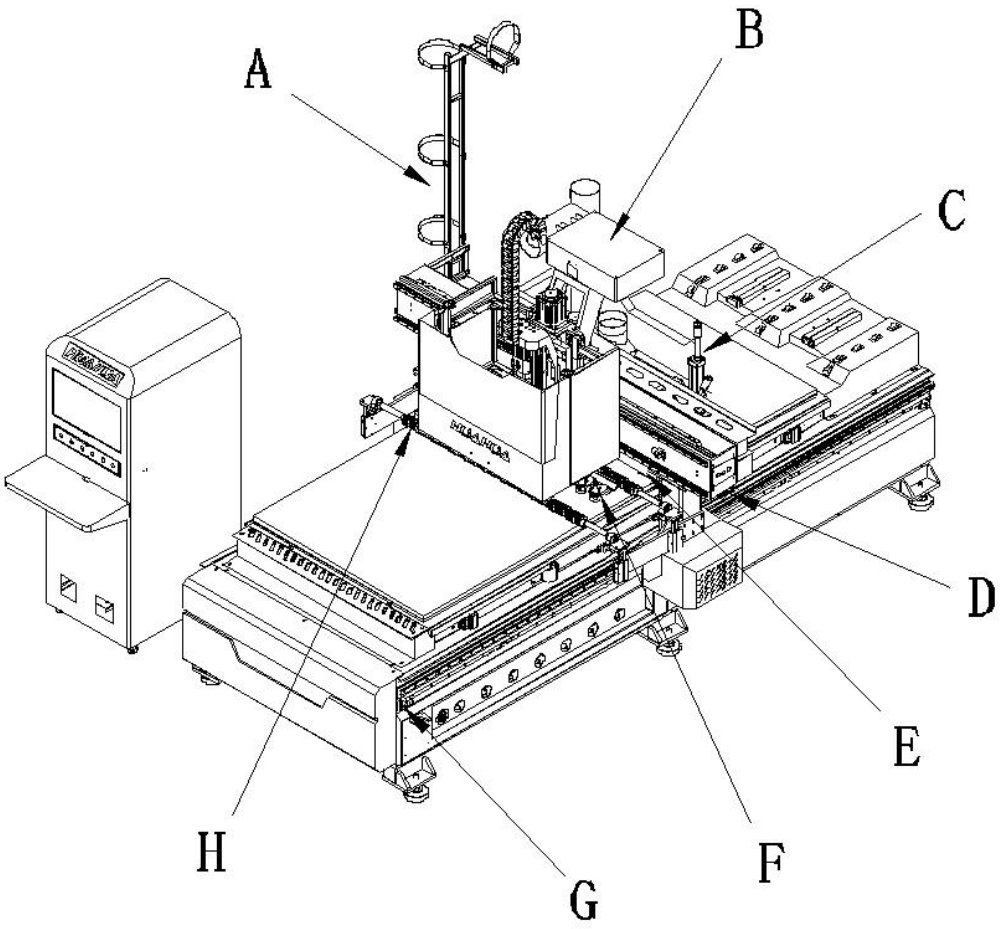
A Total Gas Source Processor

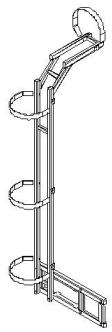
B electrical box circuit breaker

C electrical box emergency stop

D crossbeam position

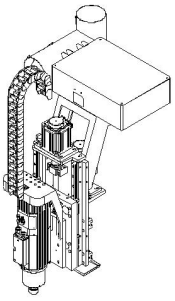
5 Introduction to working device





A. **Vacuum pipe support device**

Vacuum hose fixing device.



B. **Cutting device**

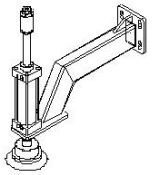
After the sheet material is lifted onto the processing table by the suction and dragging action of the feeding cylinder, the positioning cylinder positions the sheet material.

The vacuum pump adsorbs the sheet material, ensuring that the 9kW spindle in the cutting section uses the appropriate cutting tools, thus allowing for the process to proceed according to...

The processing procedure involves processing the sheet material.

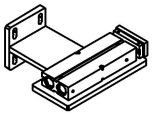
C. **Automatic loading and unloading device**

The suction cup is connected to the cylinder via a connector, and vacuum adsorption is achieved by using a solenoid valve to draw in air, thereby realizing...
The material is adsorbed and dragged for positioning and processing .



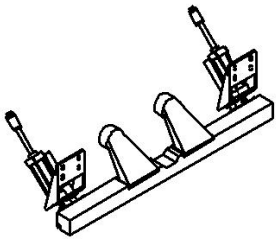
D. **Side thrust device**

Positioning is achieved by using a cylinder to push the plate from the side.



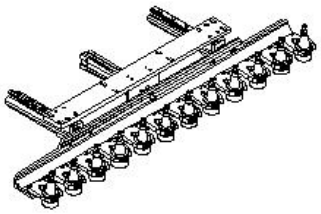
E. **Material pushing and dust collection device**

When feeding boards, it can propel the boards forward, and the device is equipped with brushes and a dust suction pipe to effectively remove dust and sawdust .



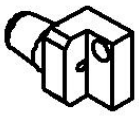
F. **Straight-line tool magazine device**

The 12 tools in the tool magazine are extended and retracted by using a cylinder to drive a guide rail slider.
The movement of the spindle enables the tool changing action.



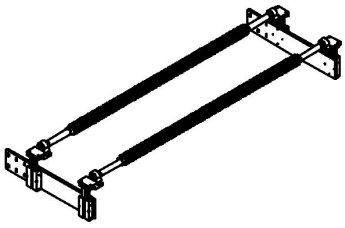
G. **Buffer and anti-collision device**

used to limit the movement of the crossbeam and the material cutting section.



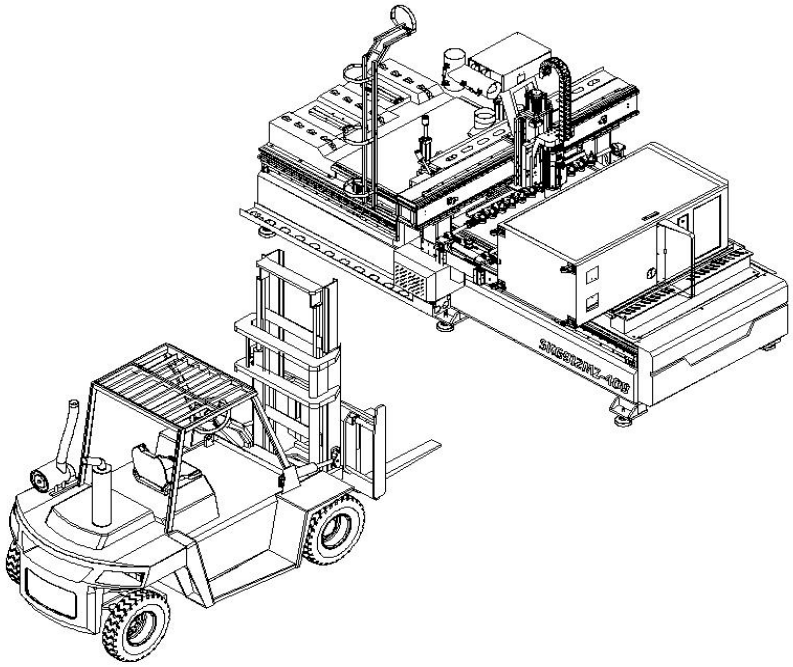
H. **Pressure roller device**

by using a solenoid valve to control the lifting and lowering of the cylinder, thus preventing the material from slipping and causing processing errors.



6 Transportation and Lifting

Equipment hoisting diagram



运输及吊装尺寸参考图

1. All movement and lifting of this machine must be performed by qualified technicians, and the operating procedures must comply with this guideline.
2. The forklift used must be of suitable capacity for the machine; the weight of the forklift should be referenced to the weight of each machine model.

1. Before lifting, ensure the forklift is positioned where the equipment's weight is evenly distributed to prevent tipping due to uneven stress.

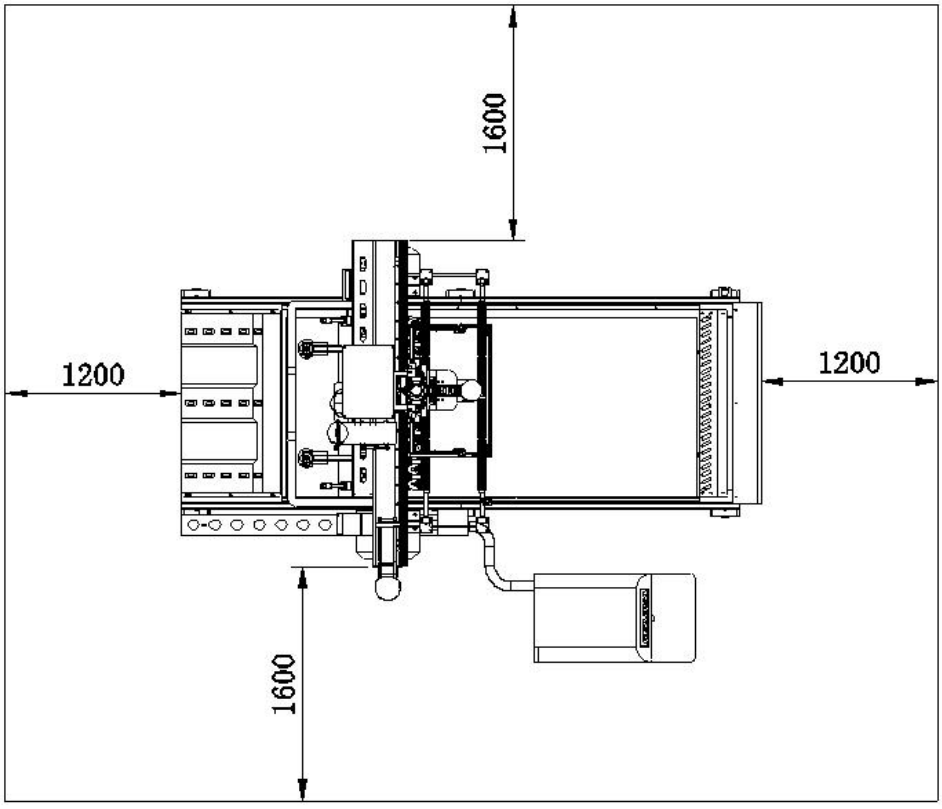
2. During equipment movement, avoid sudden movements; uneven operation may damage the machine and cause instability.

Model hoisting parameters	
model	**External dimensions (length * width * height) (meters)**
SKG 912M ZY	3.950 × 3.324 × 2.905

7 Equipment moving and installation

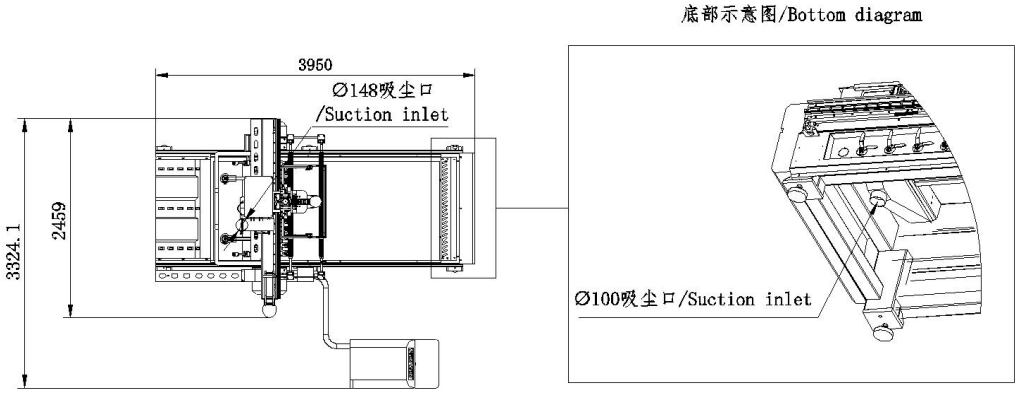
1. The machine can be moved using a forklift. The stress point should be at the bottom of the machine bed, with two stress points along the horizontal direction. When moving the machine, avoid stress on other parts outside the frame or scratches on the machine surface. Handle with care and avoid violent vibrations or collisions.

2. To protect the machine's precision and lifespan, it needs to be installed in a dry location. Avoid direct sunlight and keep it away from vibration sources. The ground should be firm and level. For convenient future maintenance and inspection, adequate space should be reserved next to the machine's installation location. Specific dimensions are shown in the diagram below:



1. Machine installation must be performed by professional technicians.
2. Before installation, ensure that all parts of the equipment have not been damaged by impact.
3. During installation, consider the working dimensions of adjacent equipment components.
4. It is recommended to maintain a certain safety distance around the equipment's working area to prevent injuries such as collisions caused by equipment operation.

8 Vacuum cleaning system conditions



The dust collection system is arranged according to the location of the equipment's dust suction port.

⚠ Poor dust extraction may harm health and affect material processing.

Vacuum pump : During use, the filter element needs to be cleaned of dust frequently and grease (German Klüber GY193 model) needs to be added , as shown in the picture below.





Add grease (German Klüber GY193 type) 1-2 times per month . Each time, press the grease gun 6-8 times ; do not add more. Adding grease while the pump is running will improve the effect. Never use non-designated grease, as the high temperature of the pump will cause the grease to melt, damaging the bearings and motor shaft, thus voiding the warranty. Clean the dust from the external filter canister and clean the filter element with an air gun every 1-3 days. Clean the sides of the pump body every 7-15 days. When cleaning the sides, clean the dust from the sides first, then remove the filter element and clean it separately. Otherwise, dust may enter the pump and damage the internal structure.

1. Do not use non-designated oil, otherwise it will damage the bearings and motor shaft.
2.Clean the side dust and filter element regularly, otherwise dust will enter the pump and damage the internal structure.

9 Electricity usage conditions

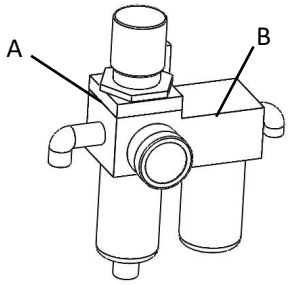
Connect the three-phase power lines from the circuit breaker , paying attention to the operating voltage and connection parameters , and whether the number of phases meets the machine's requirements. Connect the main power supply and ground the circuit.

Before starting the equipment, you need to ensure that the power grid can meet the power requirements on the nameplate.

1. Operating equipment while it is energized poses a risk of personal injury.
2. Maintenance and repairs must be performed on machines that are de-energized to avoid personal injury.

Model power parameters		
model	**Power (kW)**	**Bus wire diameter (mm²)**
SKG912MZY	20.5	6

10 Pneumatic operating conditions

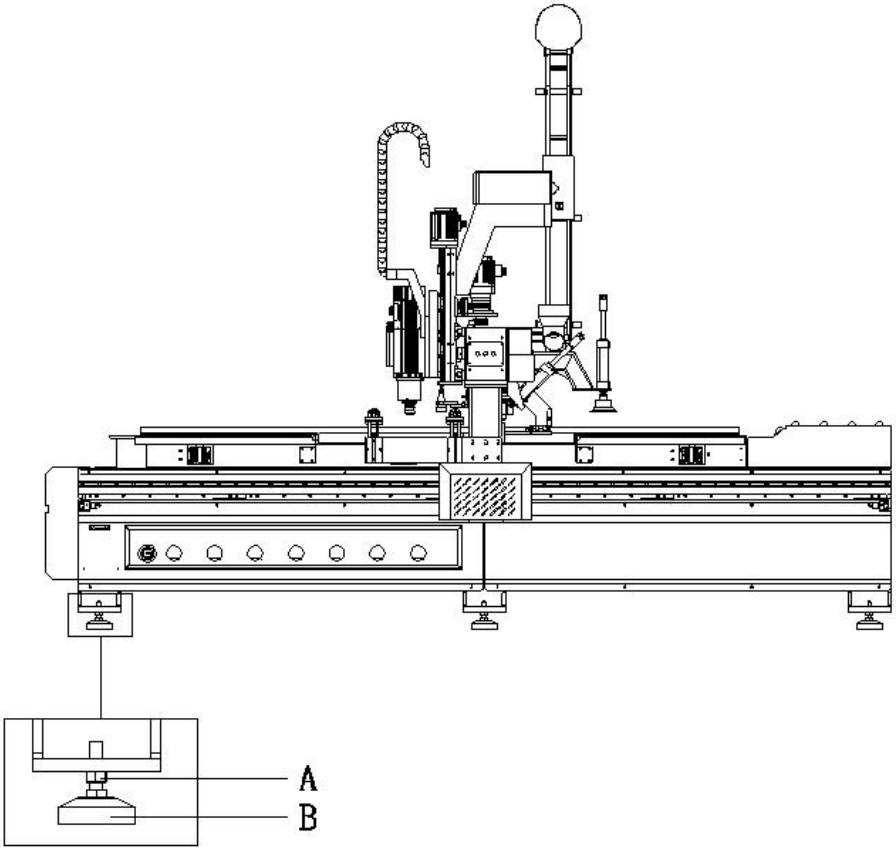


A filter B Oil Mist Cup

1. Total compressed air source pressure: 0.6-0.8 MPa.
2. Compressed air must be dried by a cooling dryer or similar equipment before being connected to the triplet unit to prevent excessive moisture from causing mechanical failure.
3. Before starting the machine each day, check the filter cup and oil mist cup of the "air source processor". The filter cup removes most of the moisture from the source air; any moisture accumulated in the cup can be expelled by pressing the yellow button at the bottom of the cup.
4. Check the oil level in the triplet unit. If it is below 1/3, add 32# anti-wear hydraulic oil to 2/3. Increase the oil level if the hydraulic oil is thick, and decrease the oil level if it is thin.

11 Installation and debugging

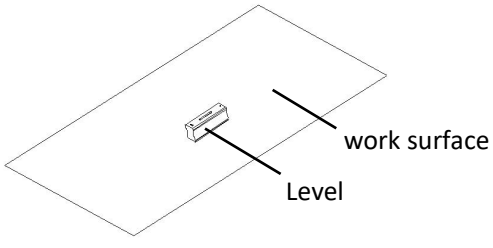
11.1 Adjust the machine level

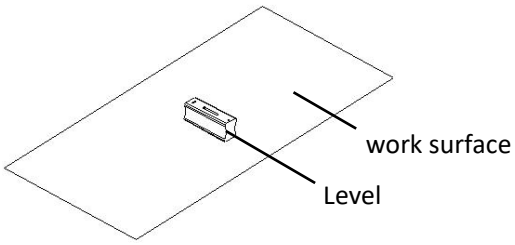


1、After the machine is stabilized, install the foot cup B onto the machine 's six support pads .

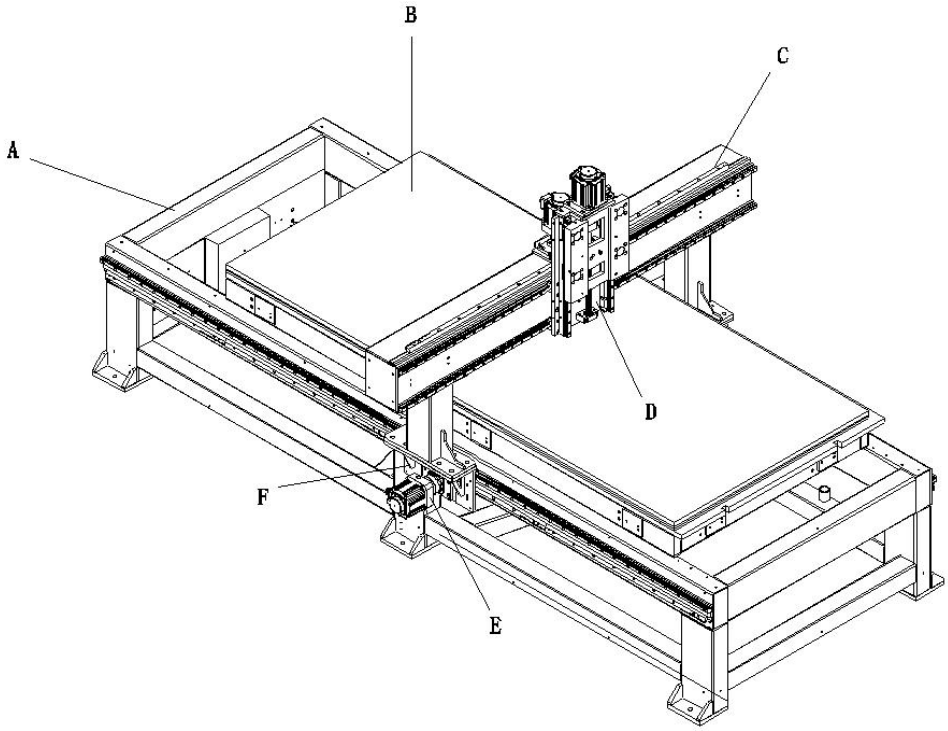
2、Clean the machine's worktable, place a level on the worktable horizontally and vertically to measure its level, and adjust the leveling screws.

Once the level indicator shows horizontal, tighten nut A to complete the machine's leveling adjustment.

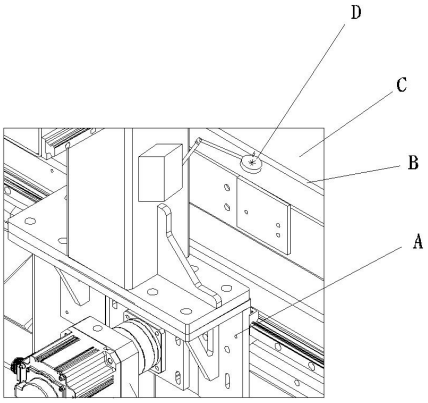




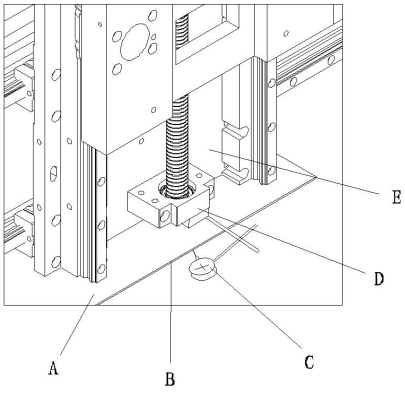
11.2 Installation and adjustment of the crossbeam



A-Frame base B-Mold plate C-Crossbeam D-Cutting base plate E-Y axis motor assembly F- Crossbeam mounting bracket



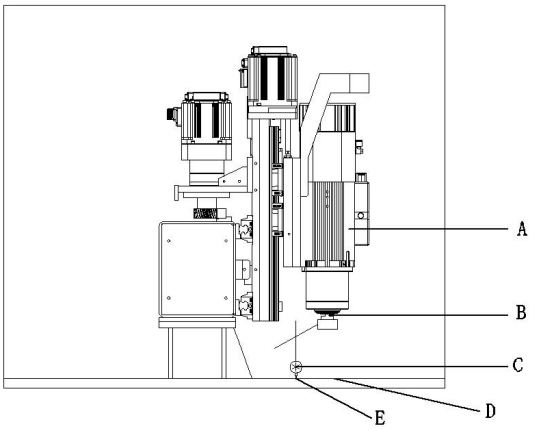
1、Place the crossbeam on the crossbeam mounting base A , screw in the screws but do not tighten them. Place the mold plate C on the machine frame table. Use a dial indicator attached to the surface of the crossbeam square tube as a reference. Adjust the pointer of the dial indicator D to contact the side B of the mold plate C. Twist the dial indicator so that its pointer points to "0". Push the column at a uniform speed to slide the crossbeam, so that the pointer swings within 0.02mm.

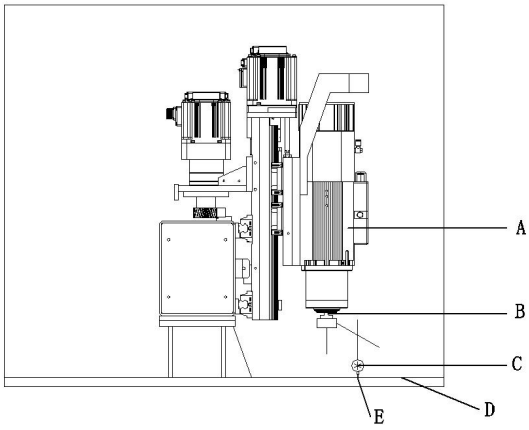


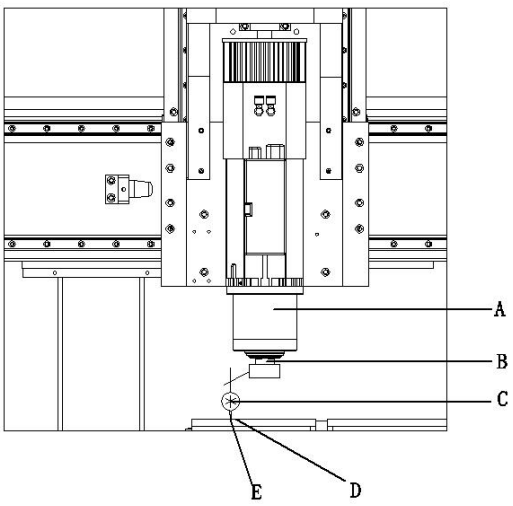
2、Using the bottom surface of the lead screw support D as a reference, attach dial indicator C to the lead screw support D. Adjust the dial indicator pointer to contact the B surface of the mold plate A, and twist the dial indicator until the pointer points to "0". Slide the cutting section E at a uniform speed, allowing the pointer to swing within 0.02~0.03mm. Finally, tighten the bolts on the crossbeam mounting base .

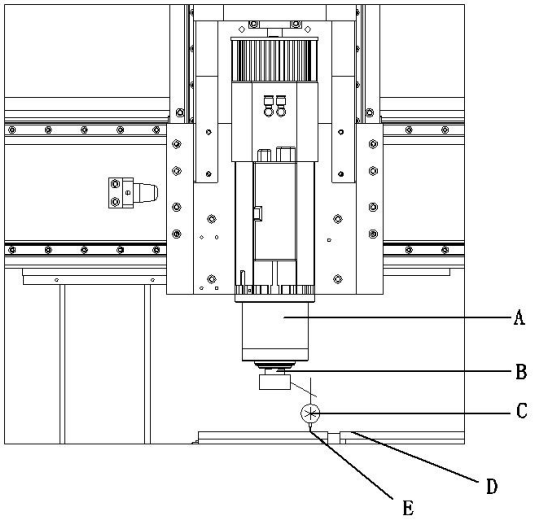
11.3 Spindle Installation and Adjustment

Use the tool holder of the spindle, locked with the special mold B, as the reference for the dial indicator. Mark a point E on the machine table D with a marker. In inch mode, turn the handwheel until the pointer of the multimeter C touches this point. Then, rotate the multimeter C to check the perpendicularity of the spindle A in all directions (as shown in the figure below). Keep the Z-axis value constant and observe the rotation value of the dial indicator. The deviation in the left and right directions should be within 0.07mm, and the deviation in the front and back directions should be within 0.1mm. If there is any deviation, adjust it by shimming copper sheets between the motor base plate and the slide plate. Repeat the adjustment until the desired result is achieved.

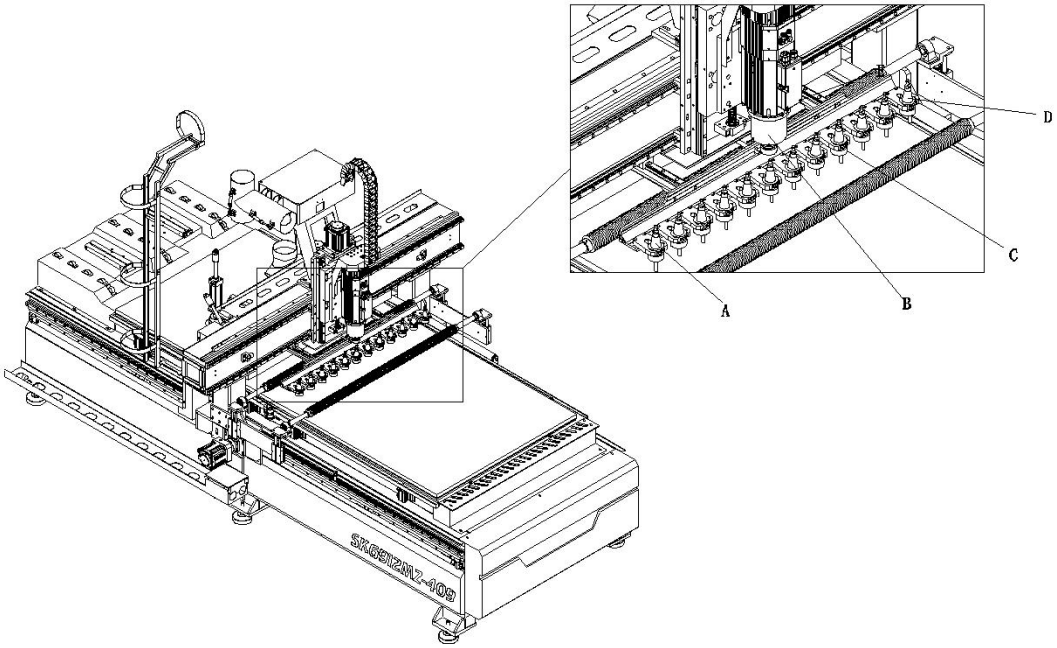




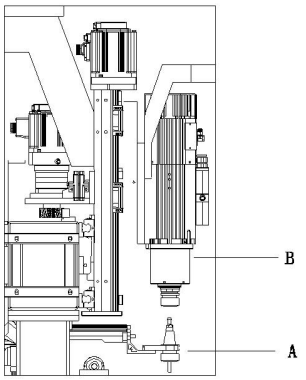




11.4 Tool magazine installation and adjustment

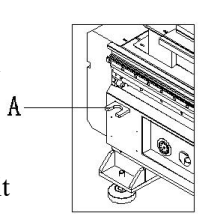


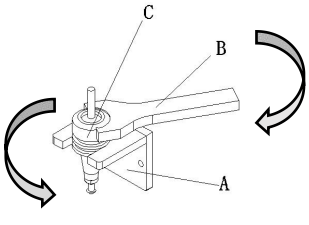
A- Knife 1 B- Spindle C- tool magazine D- Knife 12



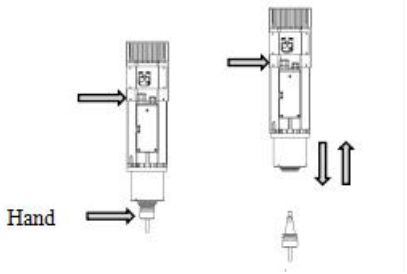
1. After installing the tool magazine components onto the crossbeam in sequence, the tool magazine needs to be adjusted. The tool magazine can be extended and retracted by a small distance by adjusting the oblong hole of the slider fixing seat. Move the spindle B up and down by handwheel to make the spindle fit with the first tool A and the last tool D respectively , so that the gaps in the front, back and left and right are basically the same, that is, adjust the tool magazine.

2. Tool Change: When it is necessary to change the tool, lock the tool C with the tool holder into place. On the automatic cutter head changer A, firmly grasp the cutter handle and turn it clockwise. Wrench B clamps the object, while turning it counterclockwise loosens it.



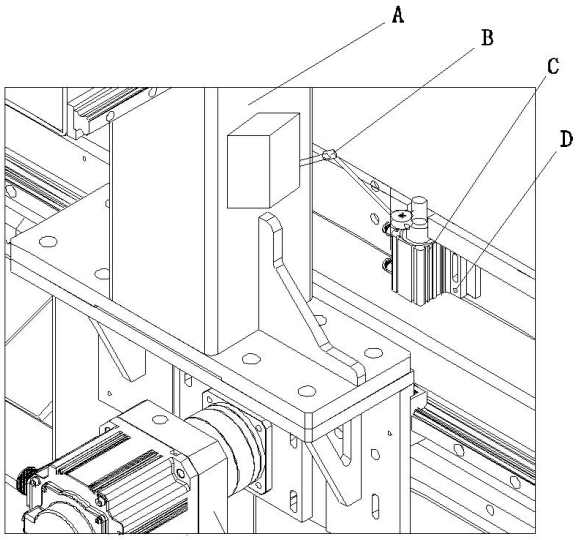


3、Spindle tool loosening and loading: When the spindle needs to be manually changed, firmly grasp the safety part of the tool holder, as shown in the right figure, press and hold the green button on the spindle, and release the button when you feel the tool drop. The tool loading and unloading steps are the reverse of the tool loosening steps

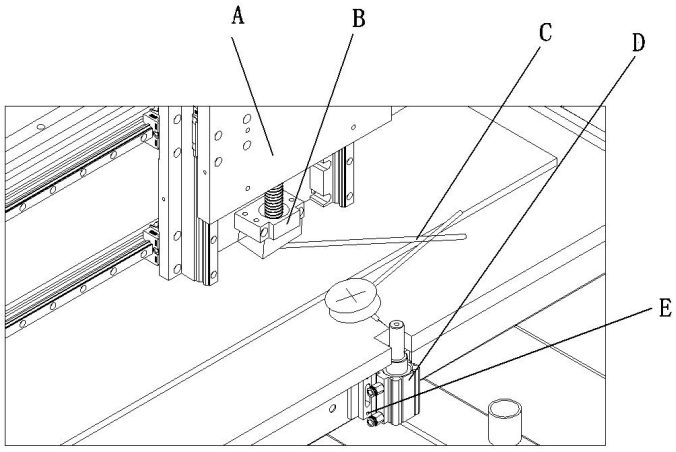


11.5 Installation and adjustment of positioning cylinder

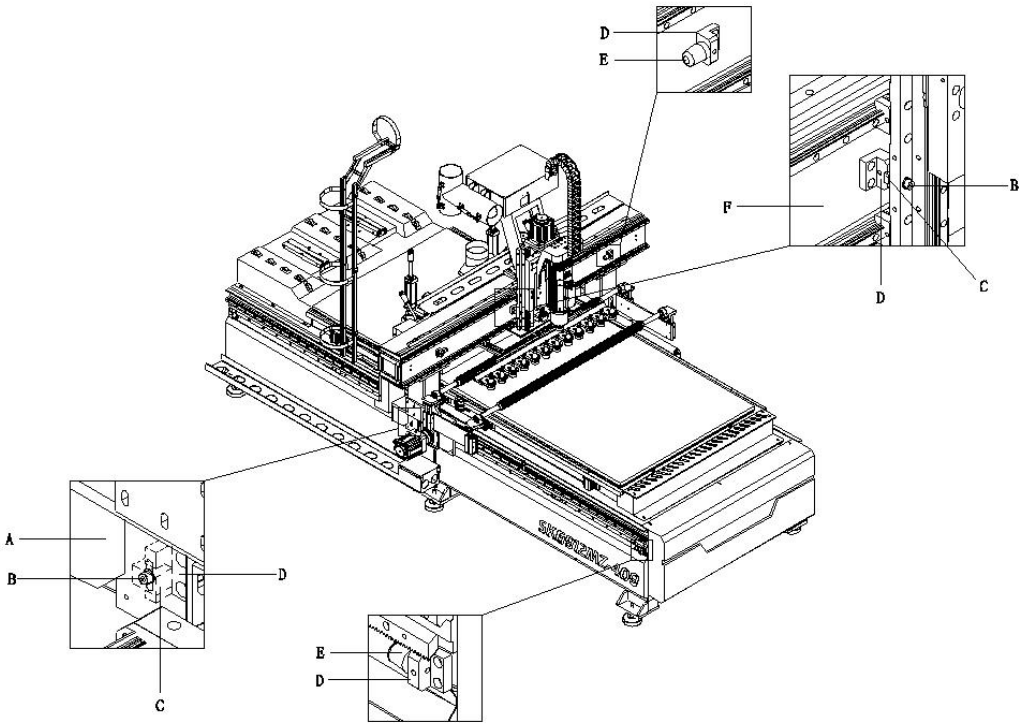
1.Side positioning cylinder: Using a dial indicator attached to the side surface A of the square tube of the crossbeam as a reference, adjust the pointer of dial indicator B to contact the side of the positioning cylinder, and rotate the dial indicator until its pointer points to "0". Move the column at a uniform speed, ensuring the pointer value is within 0.05mm when passing different positioning columns. If the deviation is too large, it can be adjusted by inserting a set screw at positioning cylinder D.



2. Using dial indicator C attached to the surface of the lead screw support B as a reference, adjust the pointer of dial indicator C to contact the side of the positioning cylinder D. Twist the dial indicator until its pointer points to "0". Move the cutting section A using the handwheel until the pointer value is within 0.05mm when passing different positioning posts. If the deviation is too large, it can be adjusted by inserting the set screw at the positioning cylinder E.



11.6 Removal of packaging fasteners



1、Remove the packaging fasteners on the crossbeam (X direction) : As shown in the figure, unscrew bolt B , remove the packaging fastener C, then remove the anti-collision base D , assemble the anti-collision base D and the cushioning rubber E together, and lock them to the corresponding installation position on the crossbeam .

2、Remove the packaging fasteners on the frame (Y direction) : Unscrew bolt B , remove packaging fastener C, then remove the anti-collision base D , assemble the anti-collision base D and the cushioning rubber E together, and lock them to the corresponding installation position on the frame.

By following the two steps above, you can remove the packaging fasteners on the machine. This will prevent the machine from being started before the packaging fasteners are removed, which could damage the machine parts. Although there may be slight differences due to model improvements, the principle remains the same.

12 Power on and power off

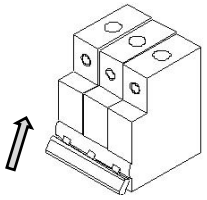
12.1 Equipment startup inspection process

1、 Check whether the air pressure and lubricating oil on the air supply triple unit are within the normal range.
2、 Check the machine for any abnormalities .
3、 Check the machine's worktable for any debris .
4、 Check that the machine's input power cord is connected correctly .

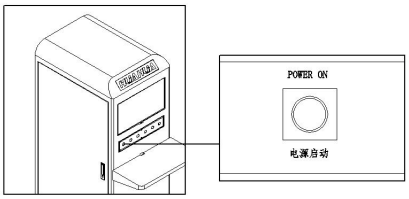
⚠ Before powering on, please strictly follow the power-on inspection procedure to avoid personal injury due to equipment malfunction.

12.2 Power on

1、After checking that there are no abnormalities in the wiring, open the electrical box and turn on the circuit breaker switch;



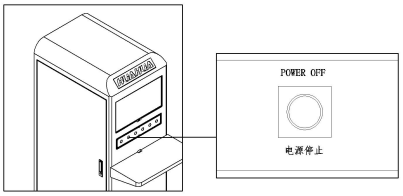
2、Then press the power button and wait for the machine to turn on.



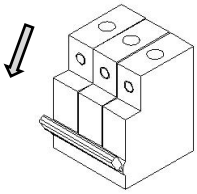
12.3 Power off

1、The machine table is free of debris, and the machine is in a safe position.

2、Press the power off button;



3、Turn off the circuit breaker switch in the electrical box.



13 Maintenance

Category	Maintenance Project	Frequency	Measure
Wear and tear	Drill bit	Once a day	Inspect the item; replace it immediately if it is damaged or severely worn.
	3A bottom cleaning knife		
	Three-blade chip-breaking straight cutter		
Lubricants	oil pump	Once a day	Check the oil level in the oil pump regularly. If the oil pump flashes an ERO signal, it indicates insufficient oil level. Add 64# hydraulic oil to 2/3 of the pump's capacity.
	vacuum pump	Once a month	Add grease 1-2 times per month (Klüber GY193 model). Each time, press the grease gun 6-8 times into each hole.
Cleaning	External filter canister for vacuum pump	Once a week	Clean the dust from the external filter canister promptly and clean the filter element with an air gun.
	Vacuum pump filter element	Once every two weeks	First, clean the dust off the sides, then remove the filter element, and then clean the filter element separately.
	Complete machine	Once a day	A thorough cleaning was performed, removing dust from inside and outside the machine body, wood chips from the tank chains, and dirt from the guide rails.
pneumatic	Triple	Once a day	Drain water promptly and check the oil level in the triplex unit regularly. If it is below 1/3, add 32# anti-wear hydraulic oil to 2/3 of the capacity.

14 List of consumable parts

Serial Number	Name	Material Code	Part	Picture	Remark
1	Sealing strip	H0308C03482	Cutting section		
2	Support rollers (right-angle combination)	H0308C00645	Feeding section		
3	cloth brush	H0308C06666	vacuuming area		
4	swivel wheels	H0308C07088	Auxiliary components		
5	Suction Cup Sleeve	H0308C00629	Suction section		

Due to factors such as product updates and upgrades, the material codes in the list of vulnerable parts may change, and the final codes shall be subject to the manufacturer's information.

15 Appendix

15.1 Random Accessories

15.2 Gas Path Diagram

15.3 Layout Diagram

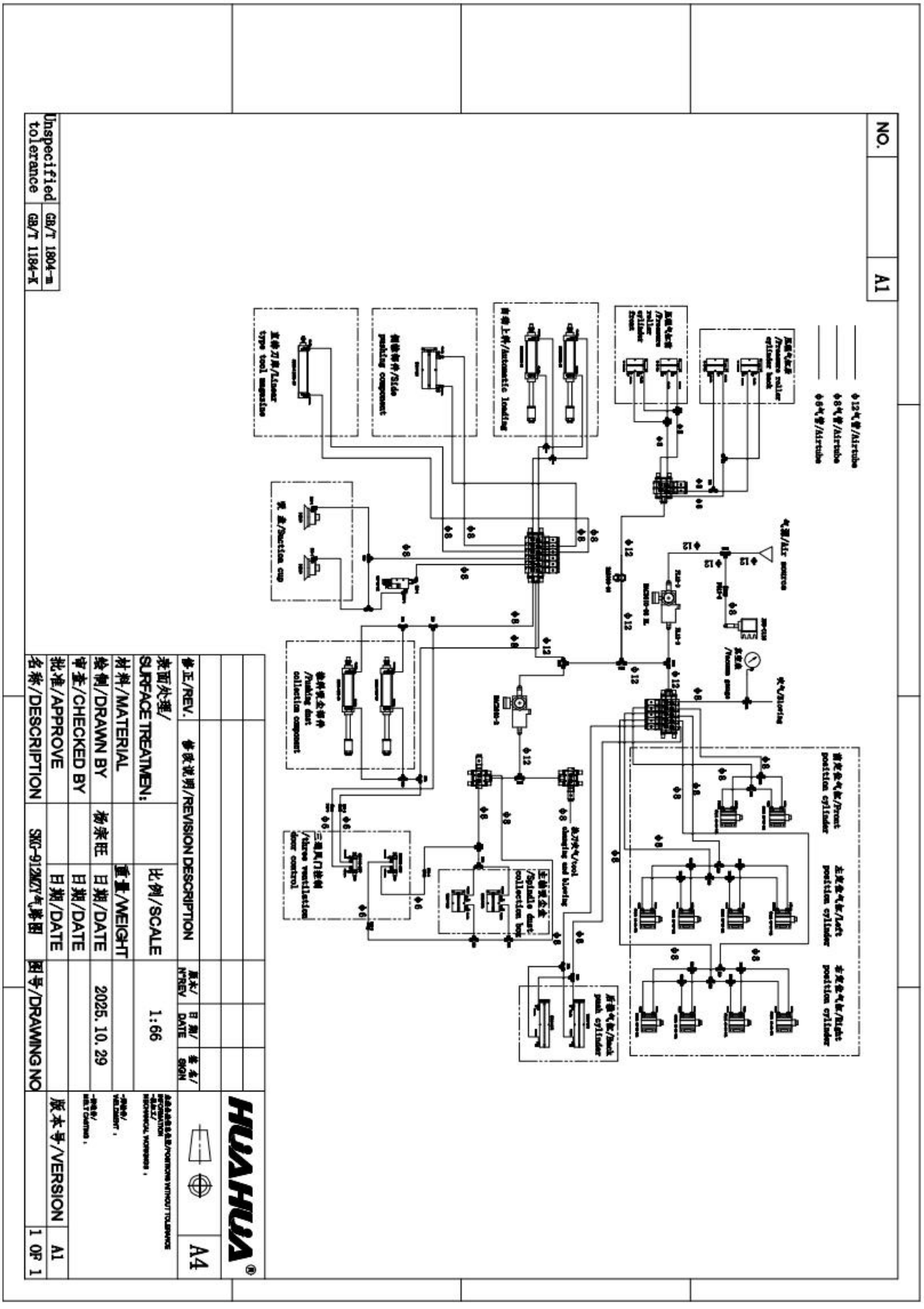
15.4 Product Certificate of Conformity

15.5 Equipment Warranty Card

Random accessories

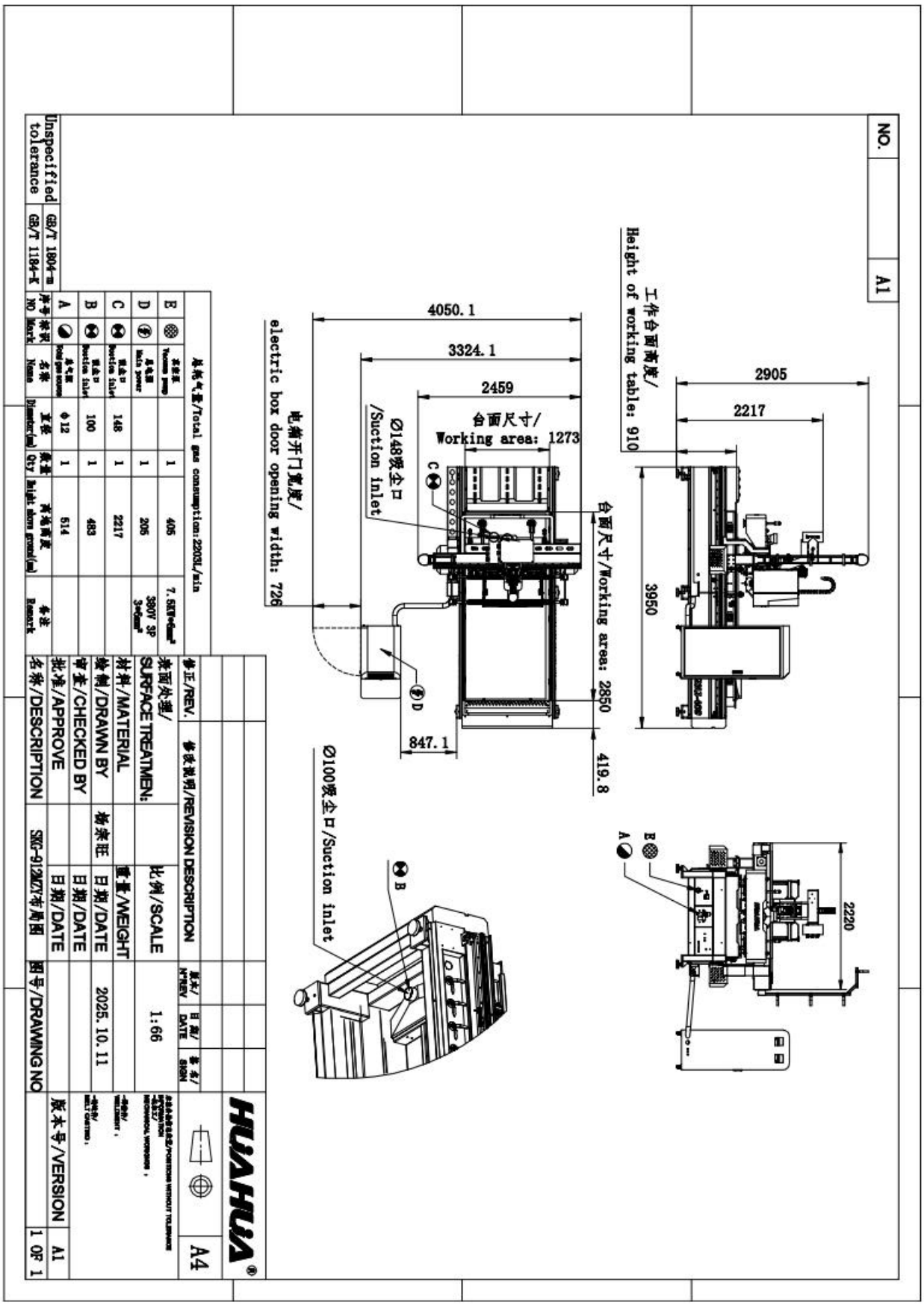
Serial Number	name	Specifications and Models	quantity
1	Imported hexagonal key		1 set
2	Open-end wrench	30-32	1
3	rigid hand	24-27	1
4	Adjustable wrench	12#	1
5	Spindle wrench	ER32 chuck	1
6	Spindle wrench	ER25 chuck	1
7	Anchor bolt washers		6
8	Anchor bolts	20	6
9	Standard drill bit	(For use with drilling machines)	6
10	Drill bit set screw	(For use with drilling machines)	20
11	Fixed knife	M8	1
12	Cleaning knife	30mm	1
13	Certificate of Conformity		1 copy
14	Toolbox		1
15	Instruction Manual		1 copy

Gas Path Diagram



广东顺德昶盛机械制造有限公司
GUANGDONG SHUNDE CHANGSHENG MACHINERY MANUFACTURING CO.,LTD.
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联结北路以南2号地块
电话：0757-27336257 网址：www.huahuamgjx.com
传真：0757-28798598 邮箱：huahua@huahuamgjx.com

Layout diagram



广东顺德昶盛机械制造有限公司
GUANGDONG SHUNDE CHANGSHENG MACHINERY MANUFACTURING CO.,LTD.
地址：广东省佛山市顺德区勒流街道新城社区中部产业园
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传真：0757-28798598 邮箱：huahua@huahuamgjx.com

HUAHUA桦桦数控®

Product Certificate

Product Name:

Product Model:

serial number:

This product has passed quality inspection and meets all quality standards and technical requirements, and is therefore permitted to leave the factory.

Inspection Manager:

Factory Manager:

Guangdong Shunde Changsheng Machinery Manufacturing Co., Ltd.

Inspection pass date: Year moon day

广东顺德昶盛机械制造有限公司
GUANGDONG SHUNDE CHANGSHENG MACHINERY MANUFACTURING CO.,LTD.
地址：广东省佛山市顺德区勒流街道新城社区中部产业园联结北路以南2号地块
电话：0757-27336257 网址：www.huahuamgjx.com
传真：0757-28798598 邮箱：huahua@huahuamgjx.com

Warranty Policy

1、 This product has a one-year warranty.

2、 During the warranty period , any malfunctions that occur under normal use in accordance with the instruction manual will be repaired free of charge.

3、 During the warranty period, repairs must be charged if any of the following occurs:

A、 Unable to provide this guarantee and valid proof of purchase .

B、Malfunctions and damage caused by misuse and improper repair .

C、Malfunctions and damages caused by the user during transportation , handling, and dropping .

D、 Other unavoidable external factors can cause malfunctions and damage .

E、Improper use can lead to damage caused by water or other solutions entering the equipment .

F、Damage caused by using power sources and voltages other than those specified in the instructions .

G、 Damage caused by improper operation of the equipment .

H、 Wear and tear on vulnerable parts .

4. The above warranties are made only, without any other express or implied warranties (including implied warranties of merchantability, reasonableness and fitness for a particular purpose), whether in contract, negligence or otherwise, and the Company shall not be liable for any special, incidental or consequential damages.

广东顺德昶盛机械制造有限公司
GUANGDONG SHUNDE CHANGSHENG MACHINERY MANUFACTURING CO.,LTD.
地址：广东省佛山市顺德区勒流街道新城社区中部产业园联结北路以南2号地块
电话：0757-27336257 网址：www.huahuamgjx.com
传真：0757-28798598 邮箱：huahua@huahuamgjx.com

appendix

Product Warranty Card

Product name _____

Product model _____

Serial No. _____

Manufacture date _____year_____month_____date

Customer name： _____

Contact： _____

Address： _____

1) **Warranty coverage：** _____ **Afterr-sales：** ____

Customer sign： _____ ☐ **good** ☐ **not good**

2) **Warranty coverage：** _____ **Afterr-sales：** ____

Customer sign： _____ ☐ **good** ☐ **not good**

3) **Warranty coverage：** _____ **Afterr-sales：** ____

Customer sign： _____ ☐ **good** ☐ **not good**

4) **Warranty coverage：** _____ **Afterr-sales：** ____

Customer sign： _____ ☐ **good** ☐ **not good**

国家高新技术企业　微信公众号　企业小程序　企业抖音号　企业视频号

广东顺德昶盛机械制造有限公司

GUANGDONG SHUNDE CHANGSHENG MACHINERY MANUFACTURING CO., LTD.

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网址(国外)/Web(foreign)：www.huahuacnc.com　邮箱(国外)/E-mail(foreign)：sales@huahuacnc.com

开料机912
电气原理图

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	Electrical schematic diagram	Cover page	页 1	Page：01/4

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Address ：Lianjie North Road, Central Industrial Park,
Leliu Street, Shunde District,Foshan City, Guangdong Province
Telephone：+86-757-27336257
Fax：+86-757-28798598

HUAHUA® 桦桦数控

技术参数/POWER DATA	
总功率/POWER ：	KW
电压/VOLTAGE ：	380VAC
总电流/CURRENT ：	A
频率/FREQUENCY ：	50HZ/60HZ
辅助电压/AUXILIARY VOLTAGE ：	220VAC
控制电压/CONTROL VOLTAGE ：	24VDC
进线电源/Incoming power supply ：	3+PE
安装标准/INSTALLATION REGULATIONS ：	GB 50057

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Electrical schematic diagram

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Electrical schematic diagram

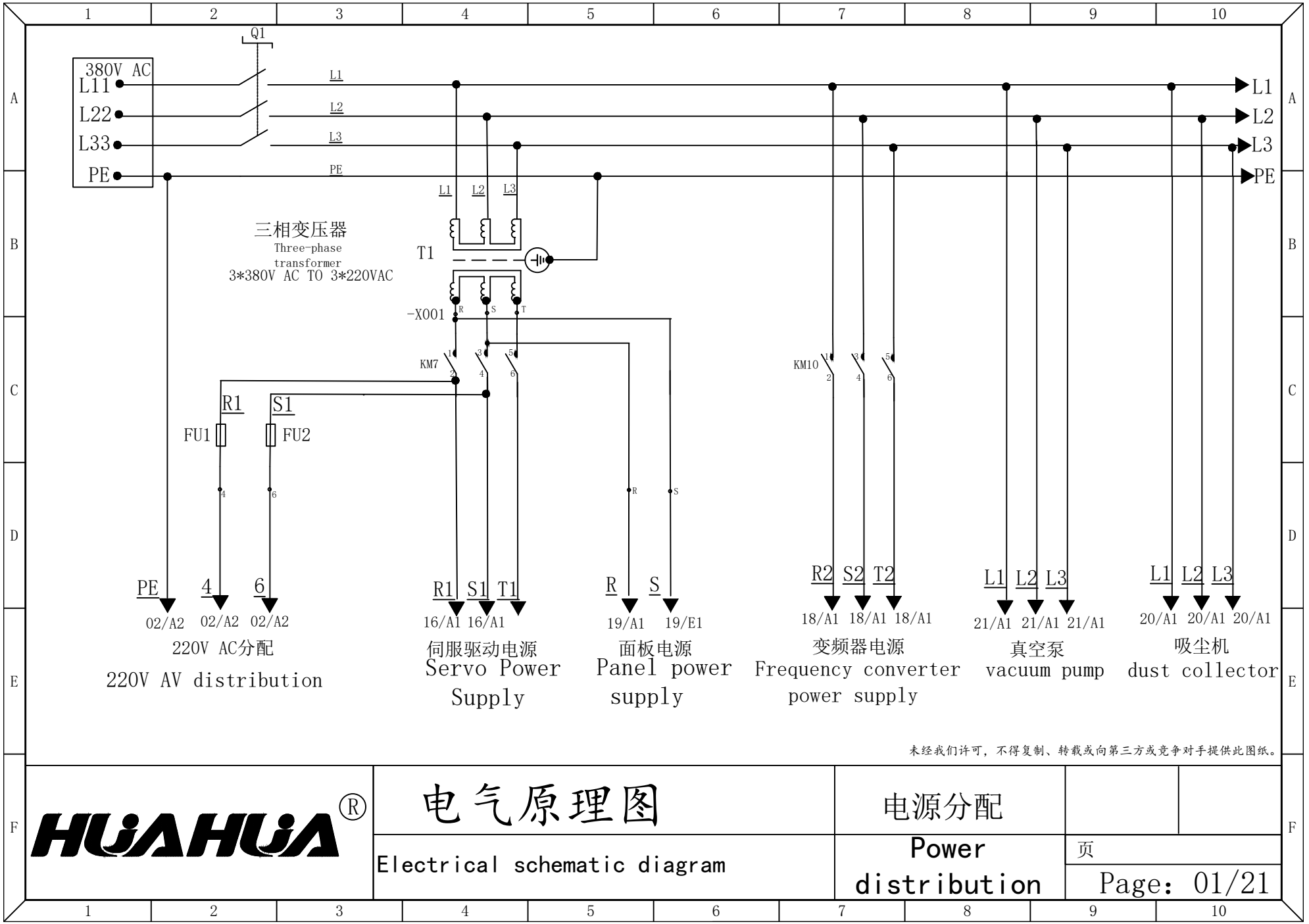
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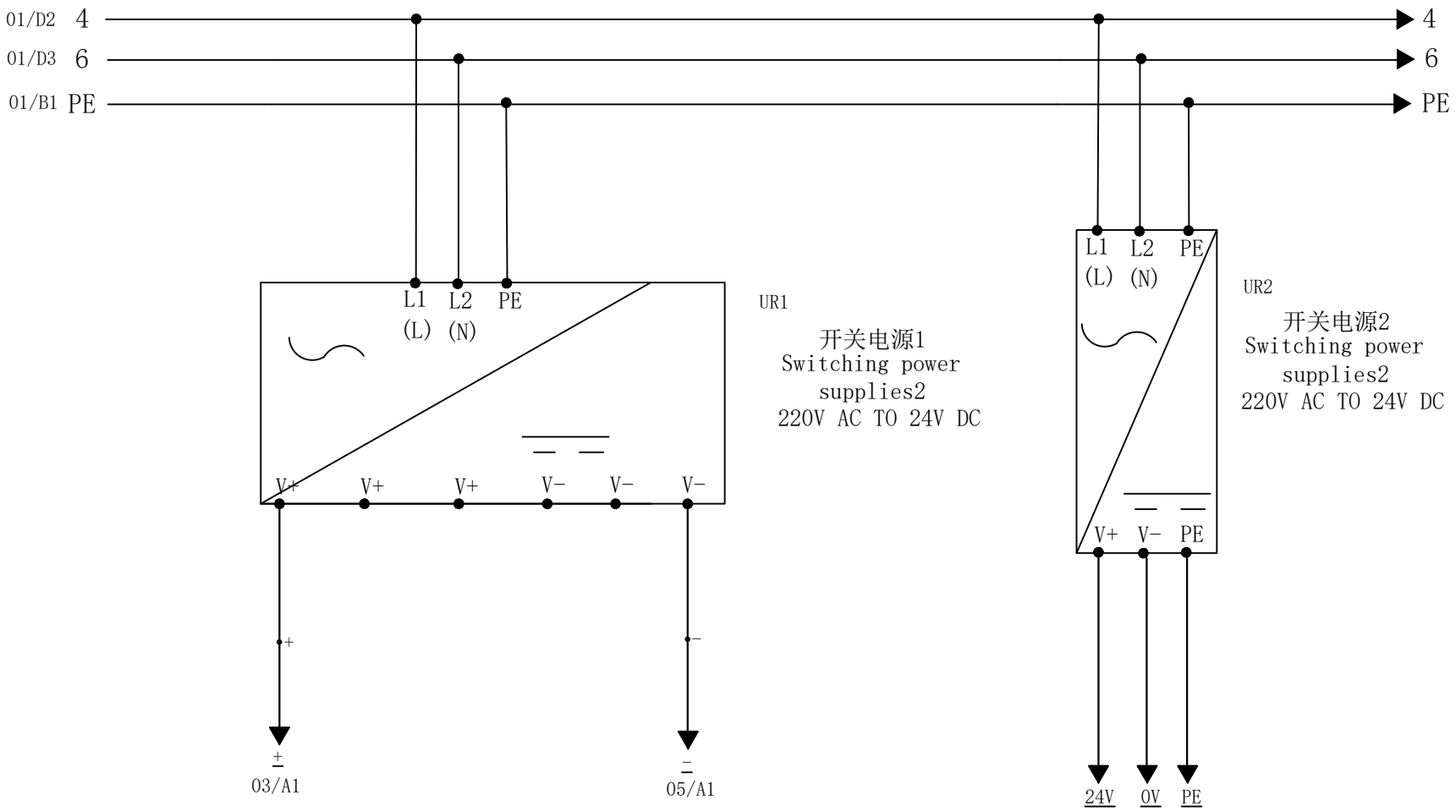
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Electrical schematic diagram

电源分配

Power distribution



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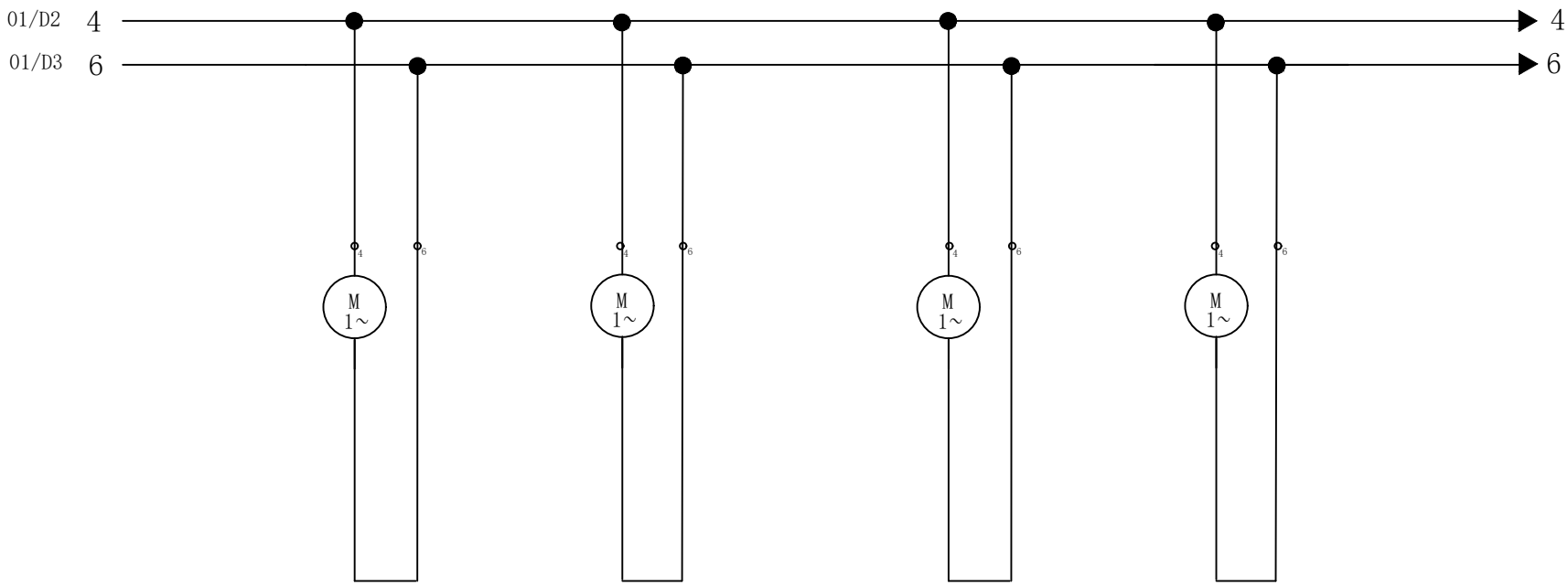
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Electrical schematic diagram

电源变换

Power conversion



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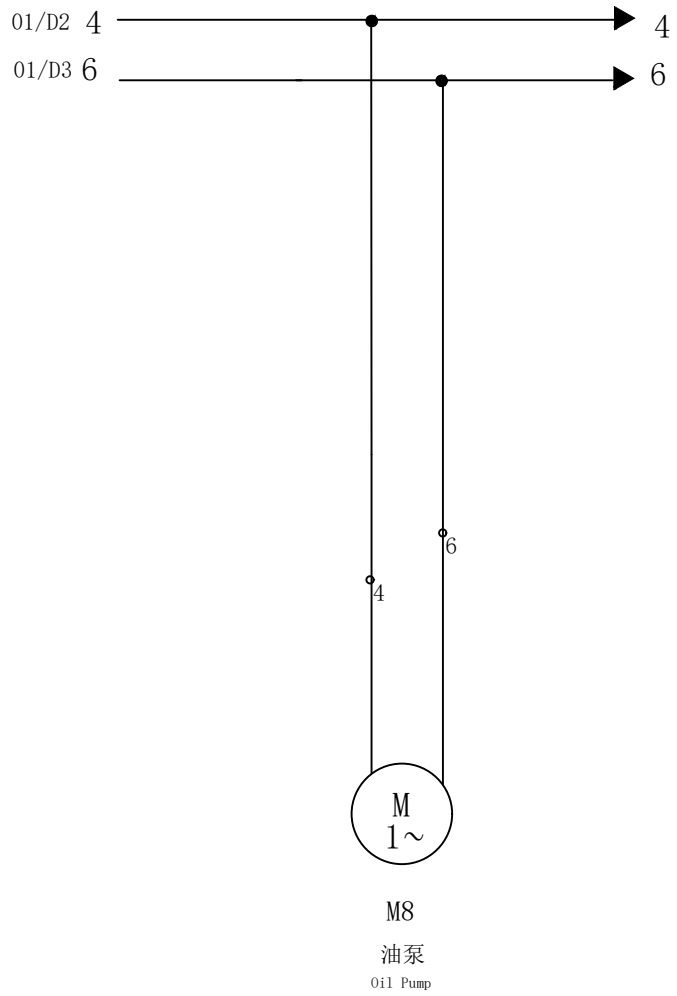
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Electrical schematic diagram

电柜通风

Cabinet ventilation



M8

油泵

Oil Pump

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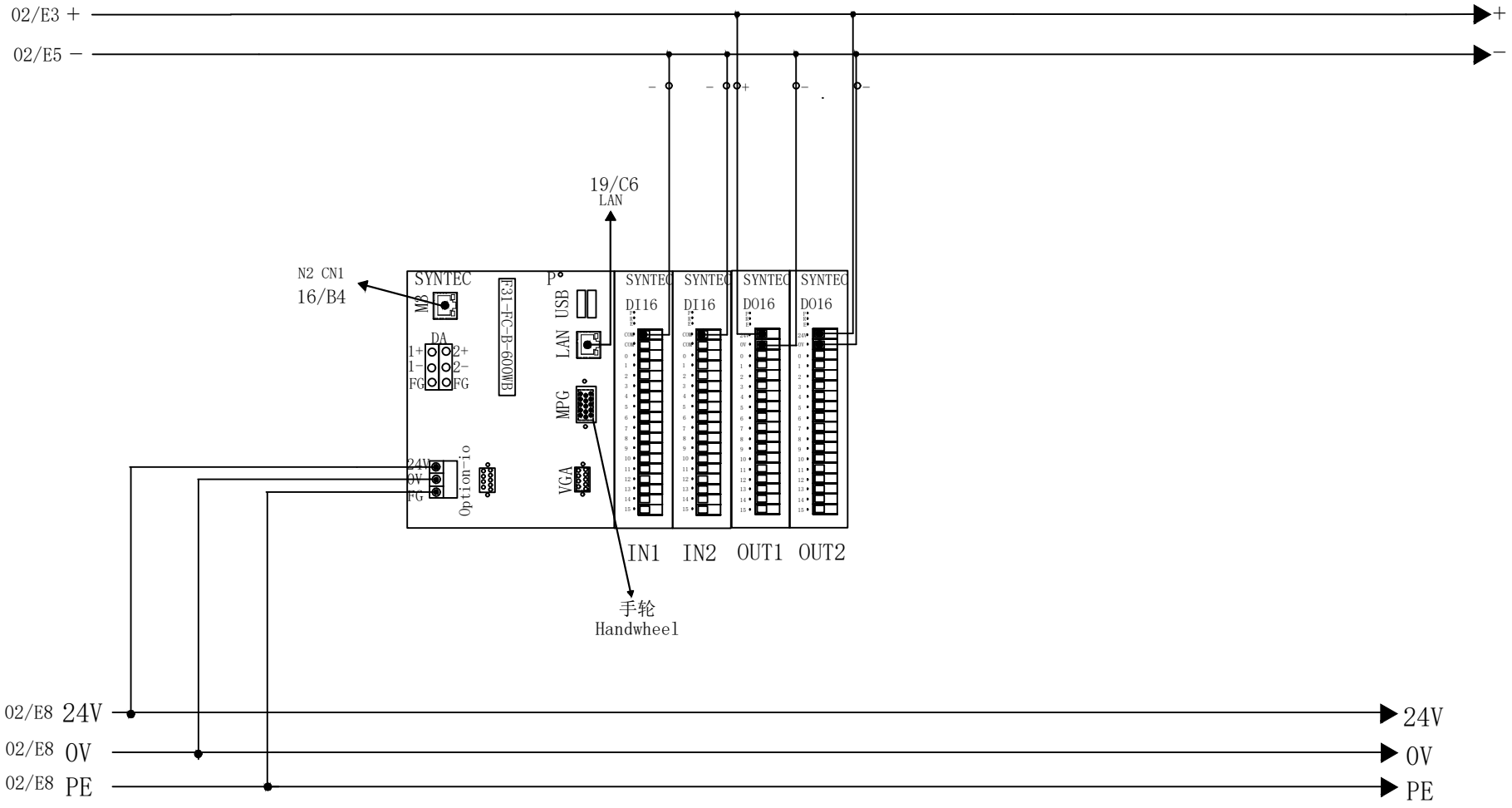
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机器润滑

Air Pressure Detection



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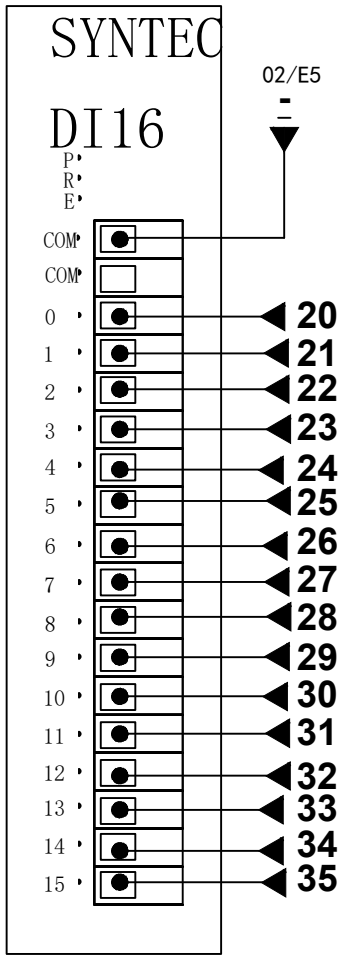
001-A01-Q1

电气原理图

Electrical schematic diagram

PLC概要

PLC Overview



Program start Button
Spindle speed stop
Emergency Stop
tool setting signal
tool setting abnormality
External box program startup
Insufficient air pressure
Inverter 1 Alarm
Tool clamping in place
Tool unclamping in place
Dust hood brush extended
Tool Magazine Pushed Out
Tool Magazine Retracted
Vacuum Pump Overload
Positive positioning 1 activated
Manual Unclamp Signal

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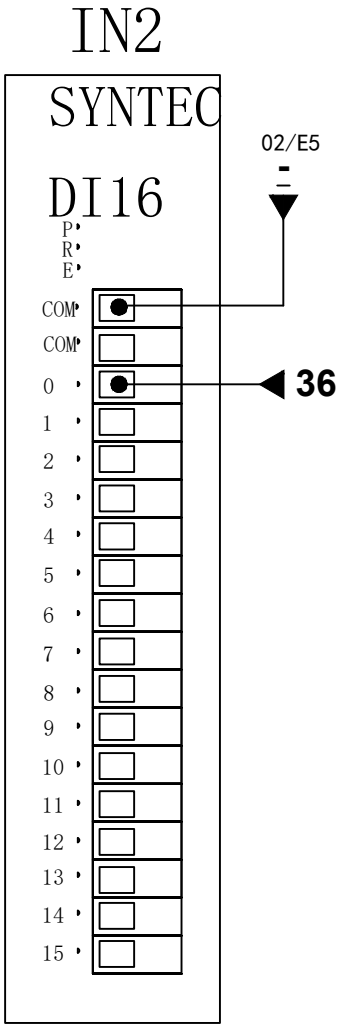
电气原理图

Electrical schematic diagram
(Electrical Cabinet 3)

PLC输入模块
PLC Input Module

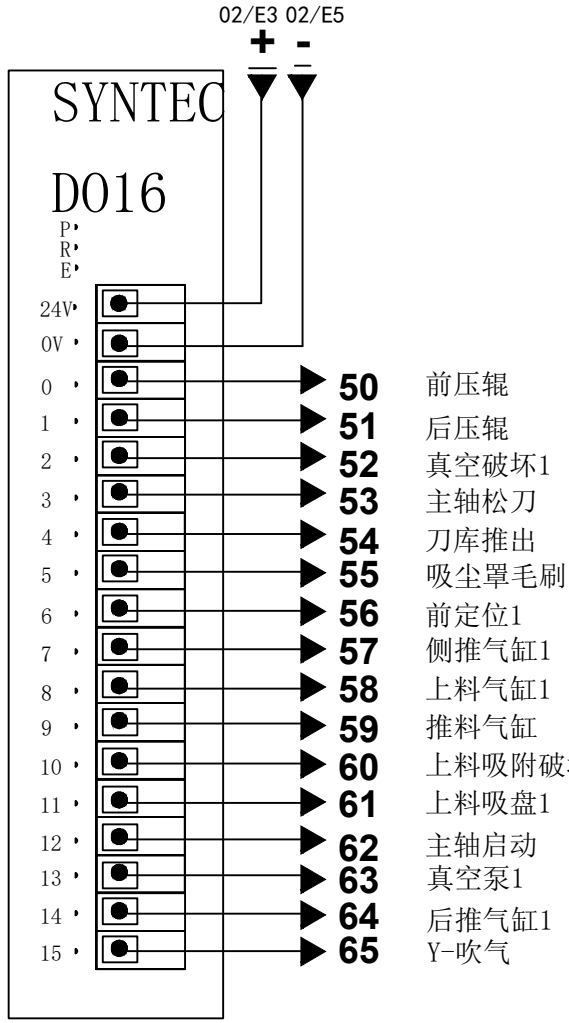
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HUAHUA®	电气原理图	PLC输入模块	=001	+A01
	Electrical schematic diagram	PLC Input Module	页 Page: 06/21	



50	前压辊	Front pressure roller
51	后压辊	Rear pressure roller
52	真空破坏1	Vacuum Break 1
53	主轴松刀	Spindle Unclamping
54	刀库推出	Tool Magazine Extension
55	吸尘罩毛刷	Dust hood brush
56	前定位1	Front positioning 1
57	侧推气缸1	Side-push cylinder 1
58	上料气缸1	Feeding cylinder 1
59	推料气缸	Pushing cylinder
60	上料吸附破坏1	Feeding adsorption failure 1
61	上料吸盘1	Feeding suction cup 1
62	主轴启动	Spindle Start
63	真空泵1	Vacuum Pump 1
64	后推气缸1	Rear-push cylinder 1
65	Y-吹气	Y-blow

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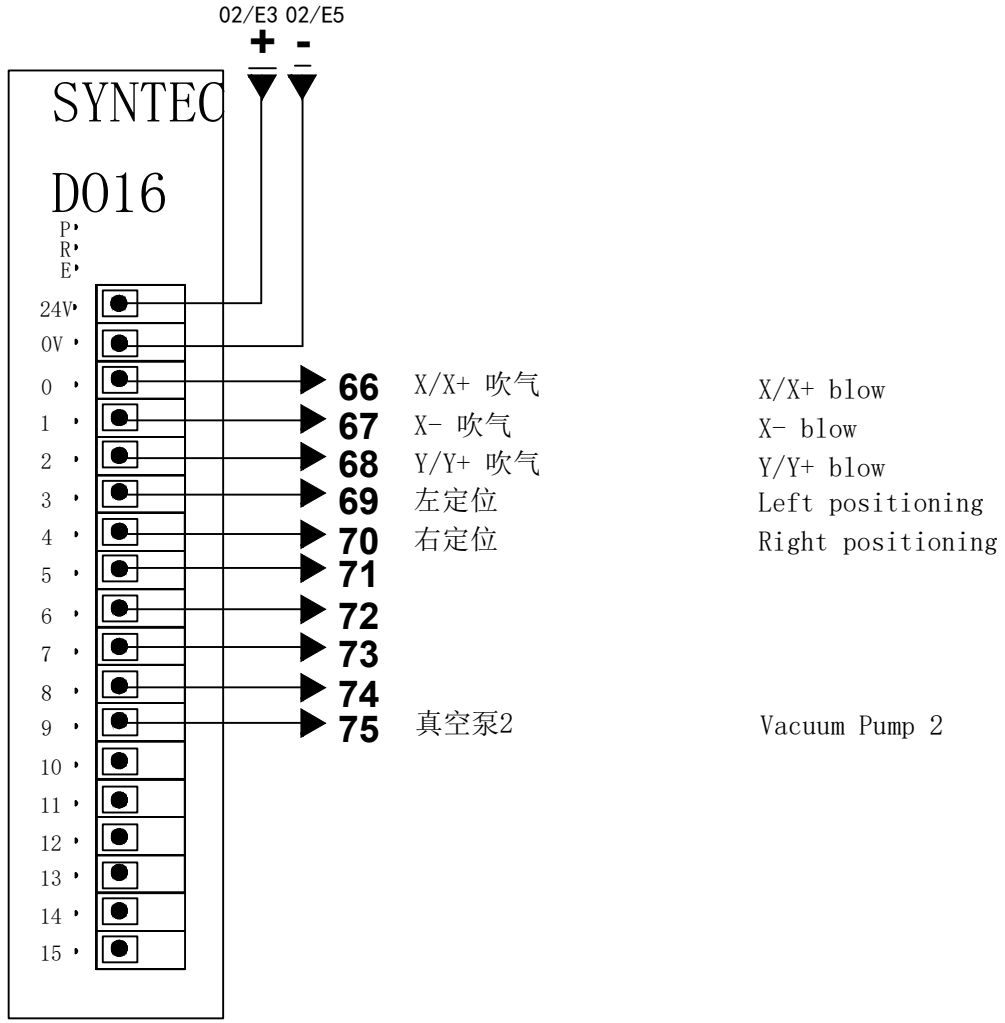
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电气原理图

Electrical schematic diagram

PLC输出模块

PLC Output Module



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电气原理图

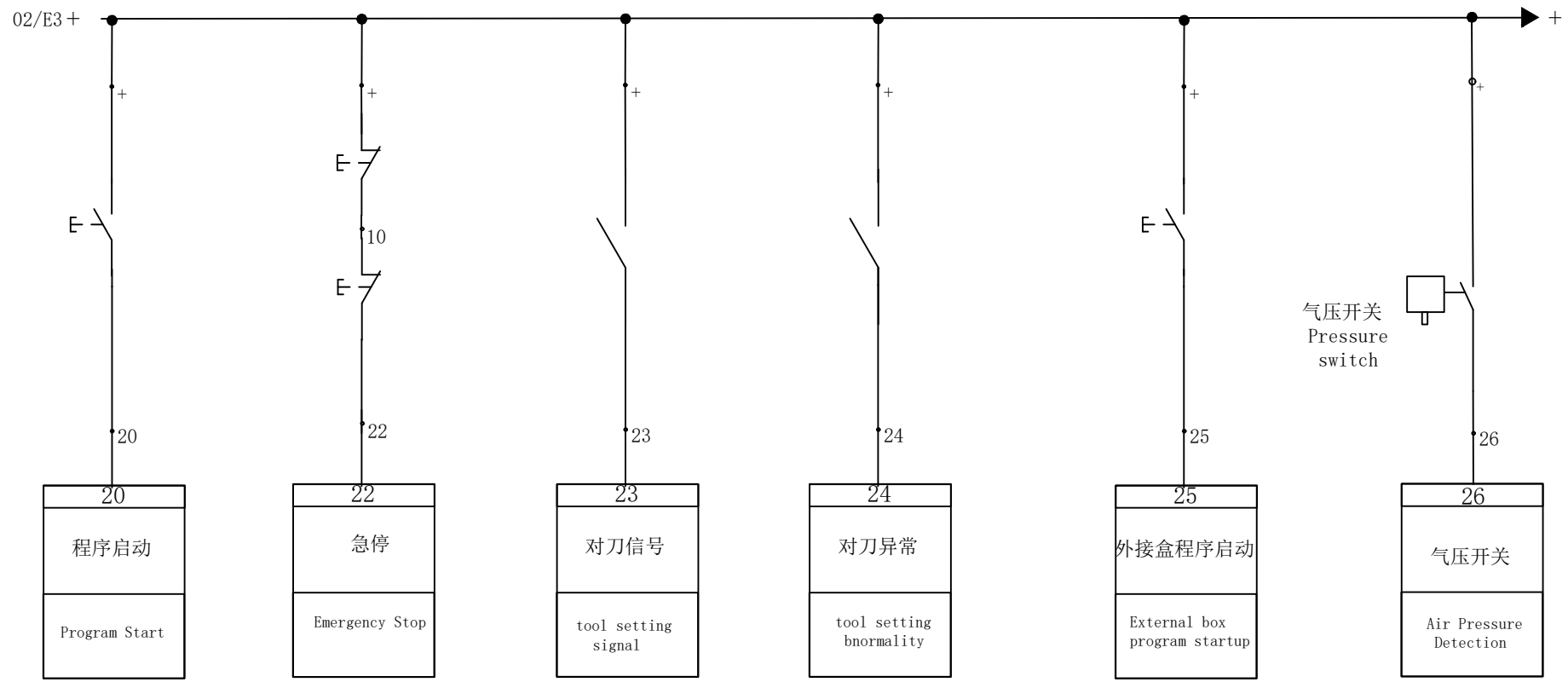
Electrical schematic diagram

PLC输出模块

PLC Output Module

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电气原理图

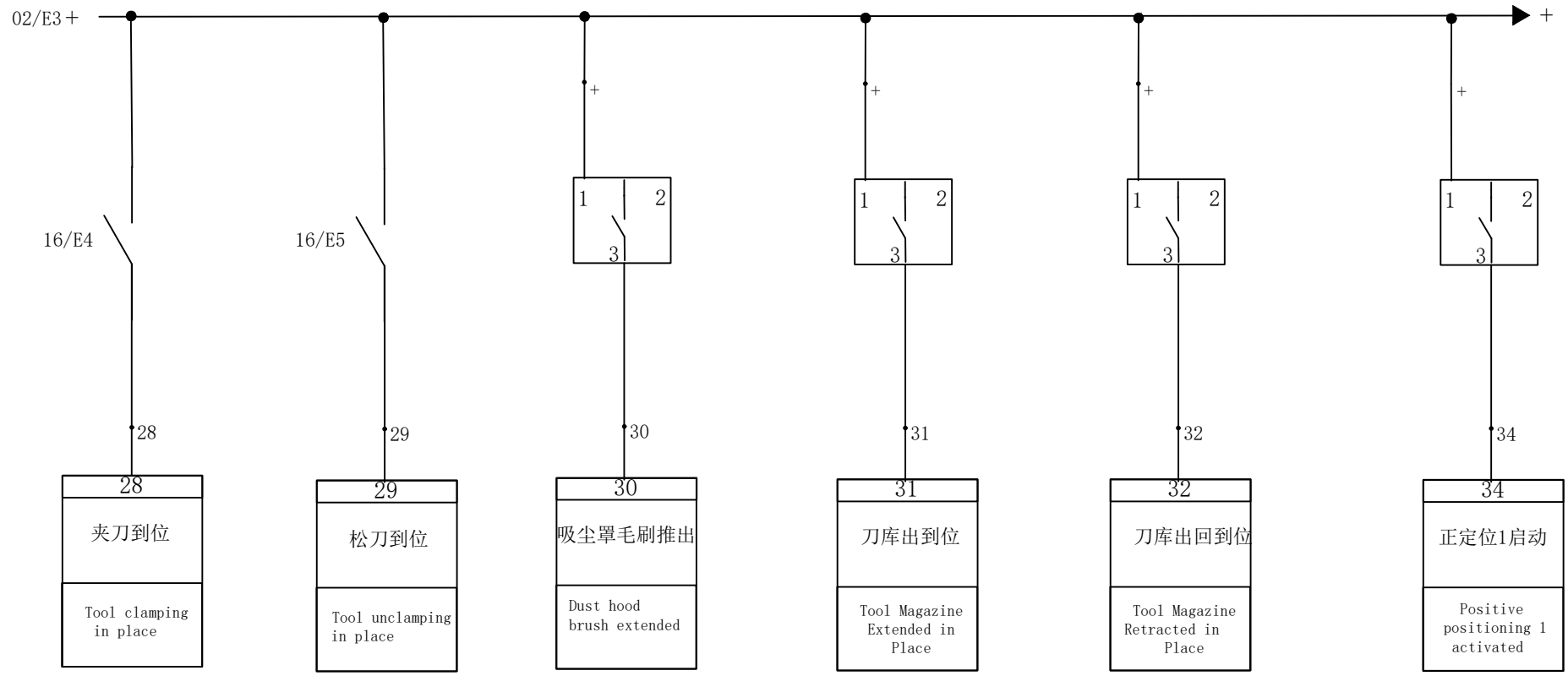
Electrical schematic diagram

PLC输入

PLC Input

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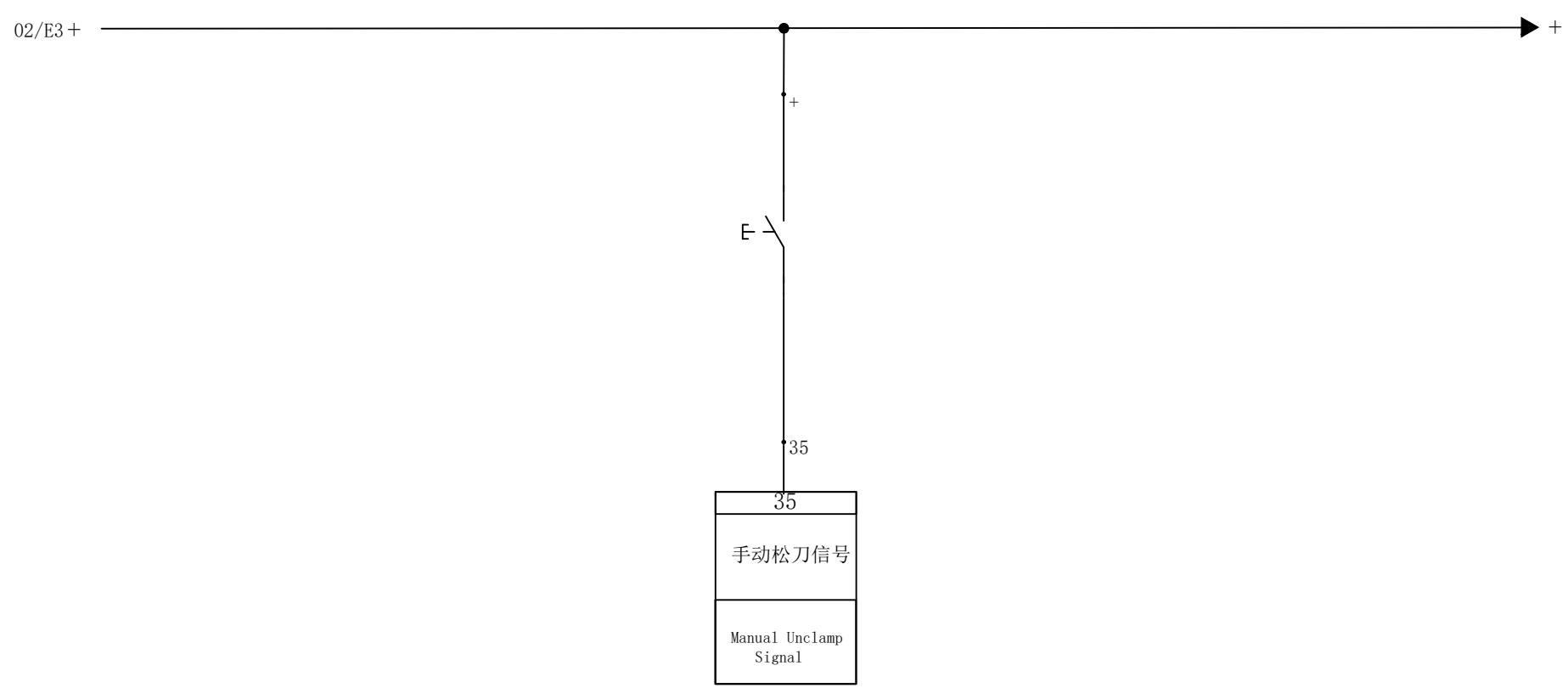
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电气原理图

Electrical schematic diagram

PLC 输入

PLC Input



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电气原理图

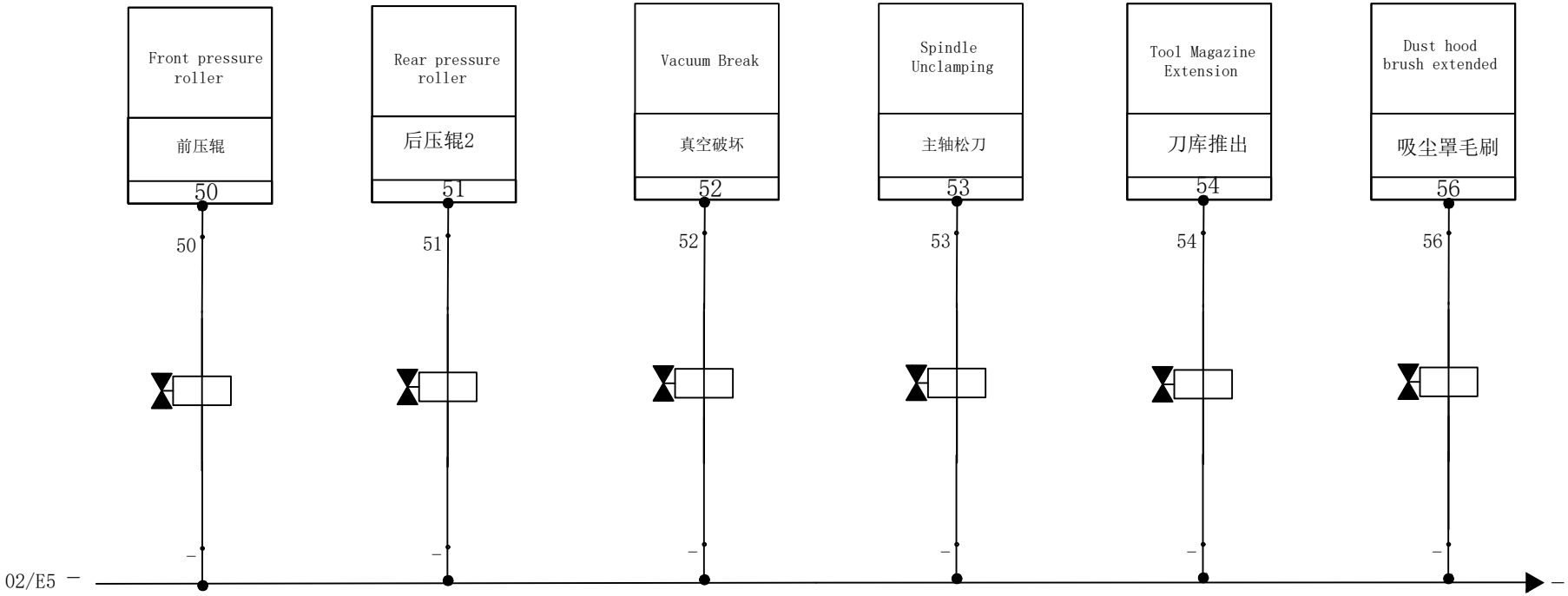
Electrical schematic diagram

PLC 输入

PLC Input

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电气原理图

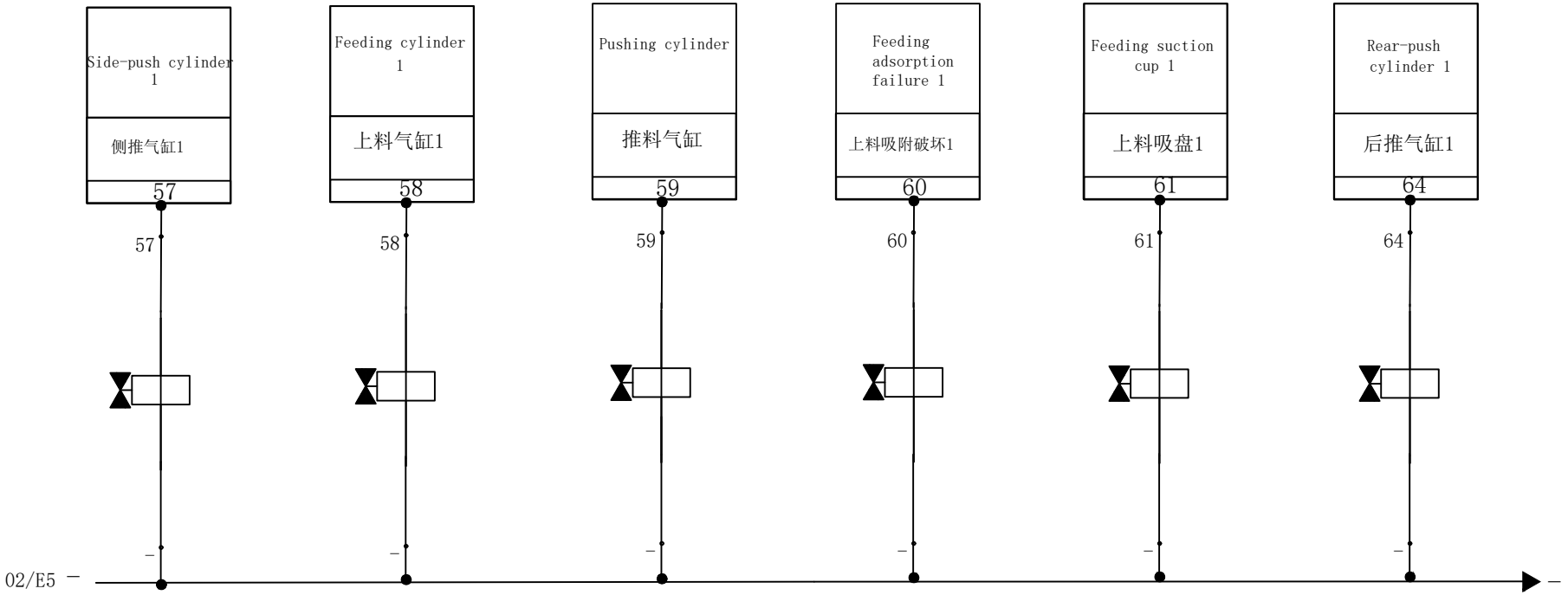
Electrical schematic diagram

PLC 输出

PLC Output

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电气原理图

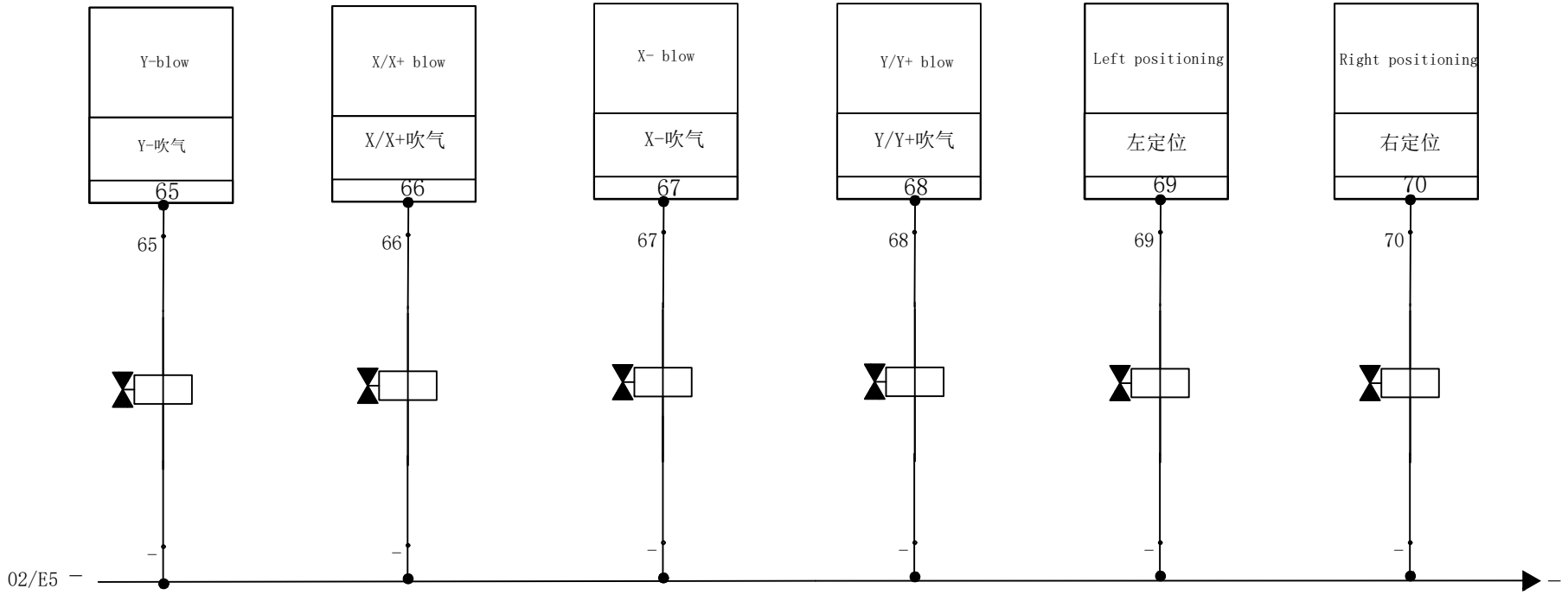
Electrical schematic diagram

PLC 输出

PLC Output

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电气原理图

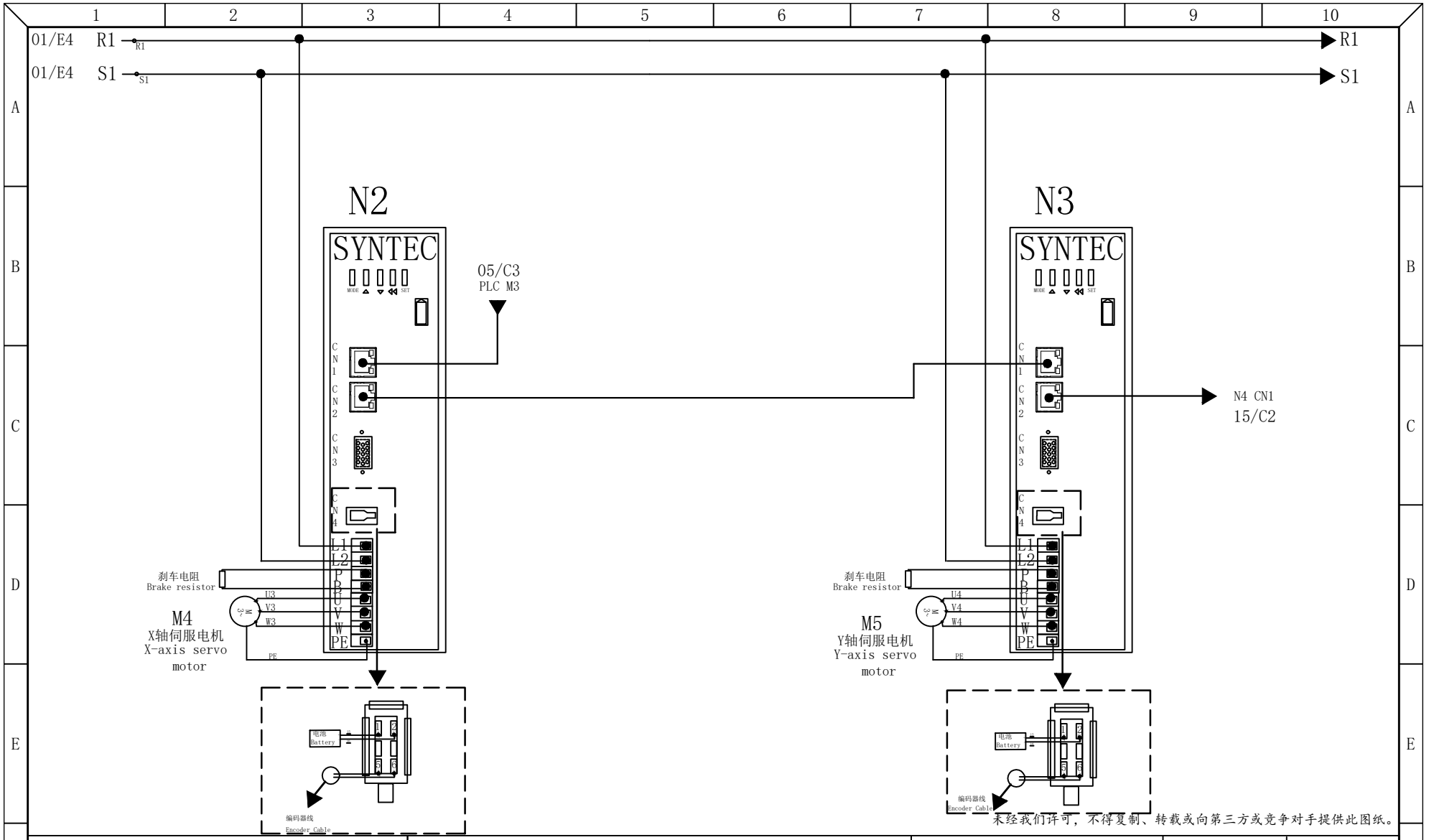
Electrical schematic diagram

PLC 输出

PLC Output

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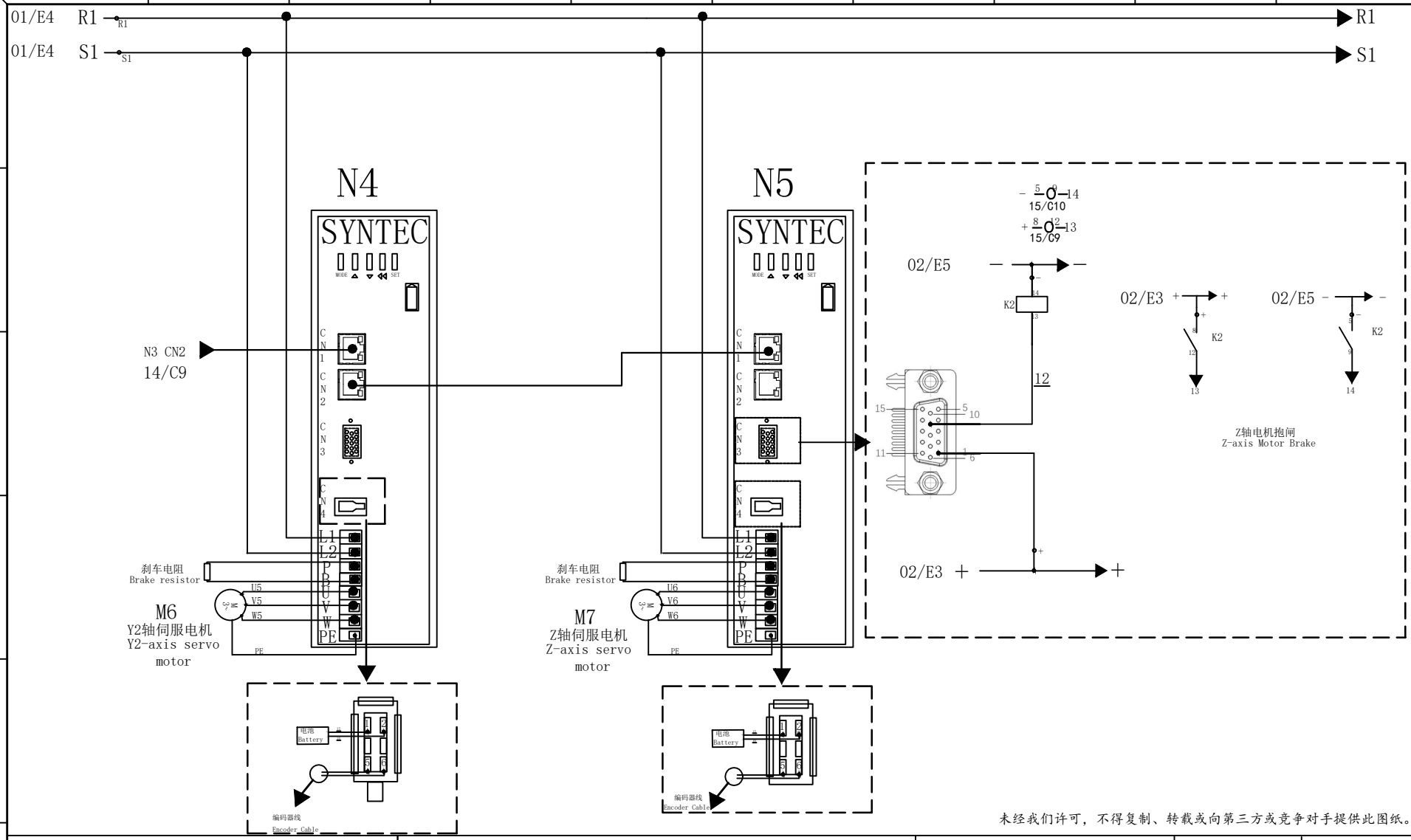
电气原理图-电柜3

Electrical schematic diagram

(Electrical Cabinet 3)

伺服驱动

Servo Drive



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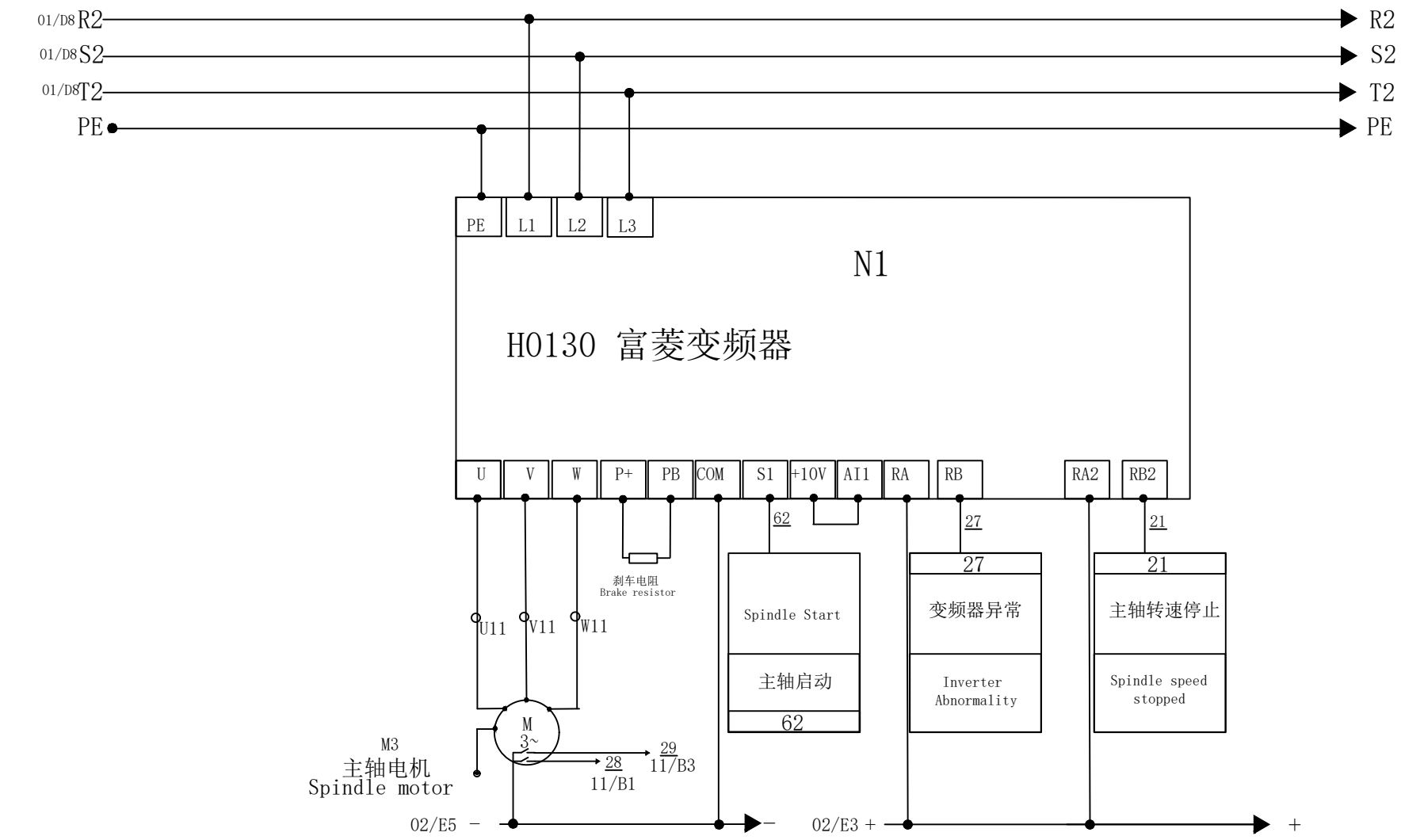
电气原理图-电柜3

Electrical schematic diagram

(Electrical Cabinet 3)

伺服驱动

Servo Drive



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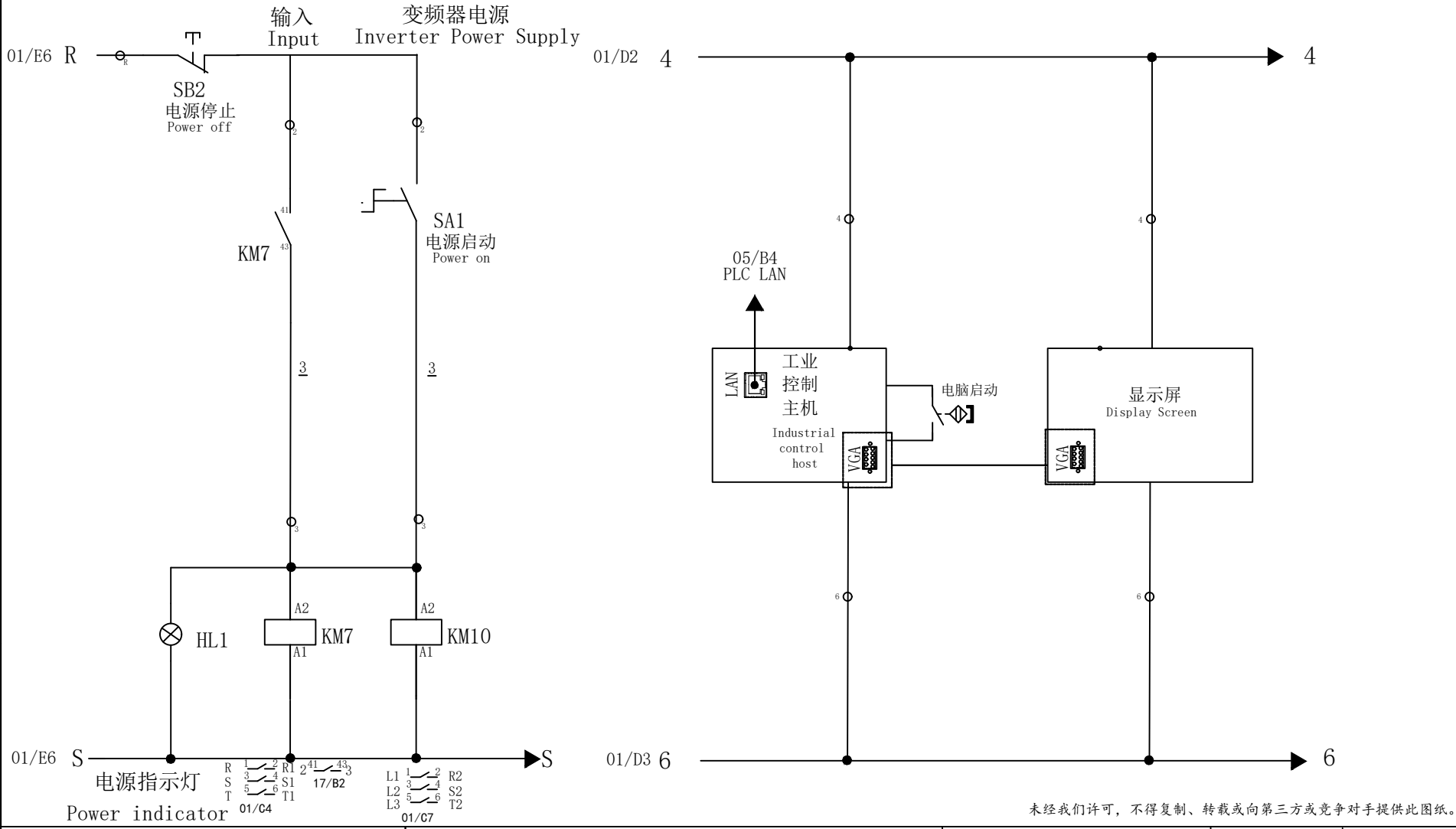
电气原理图 -电柜2

Electrical schematic diagram

(Electrical Cabinet 2)

主轴电机部分

Spindle motor Part



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电气原理图

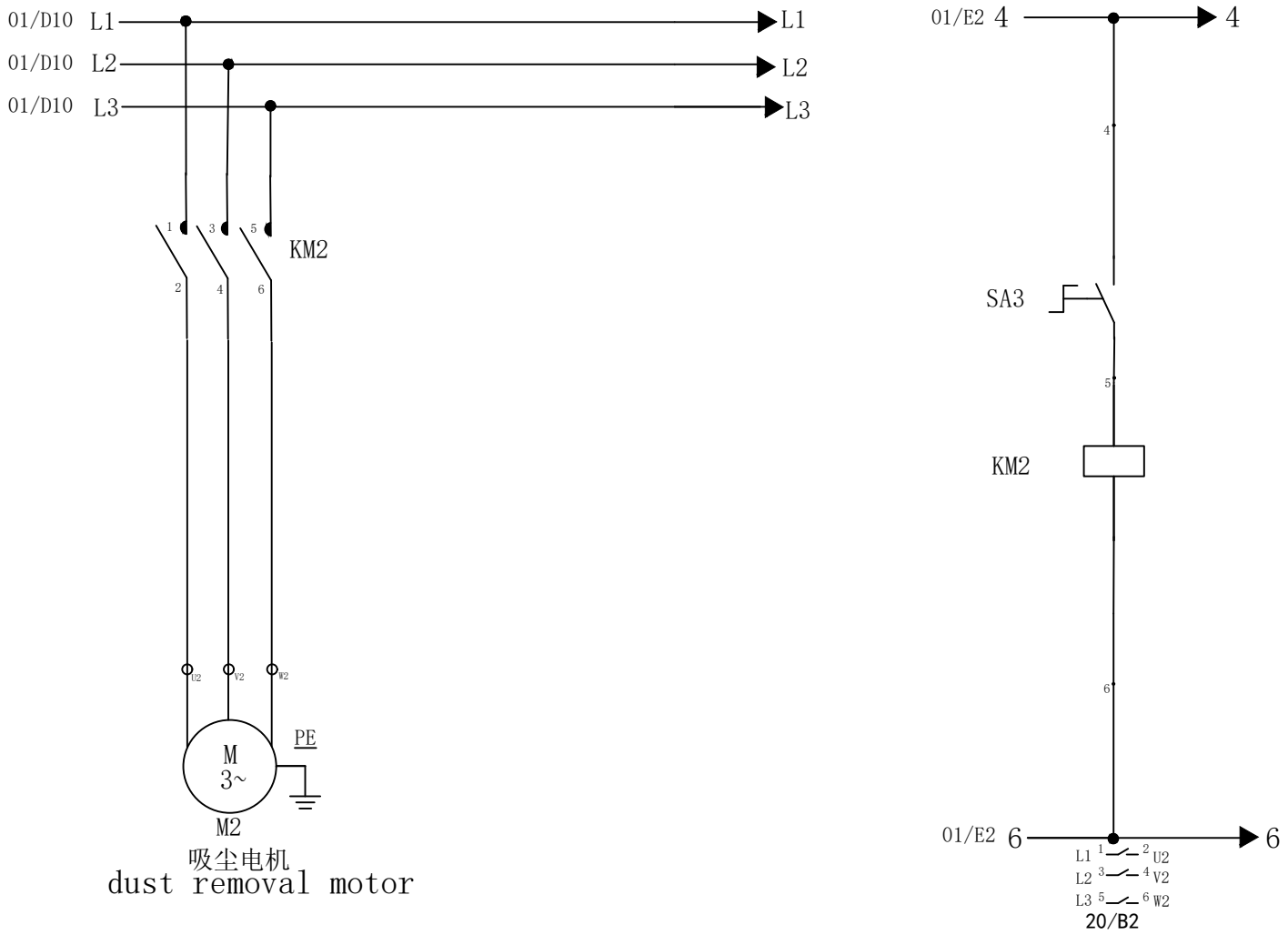
Electrical schematic diagram

控制面板回路

Control panel loops

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电气原理图

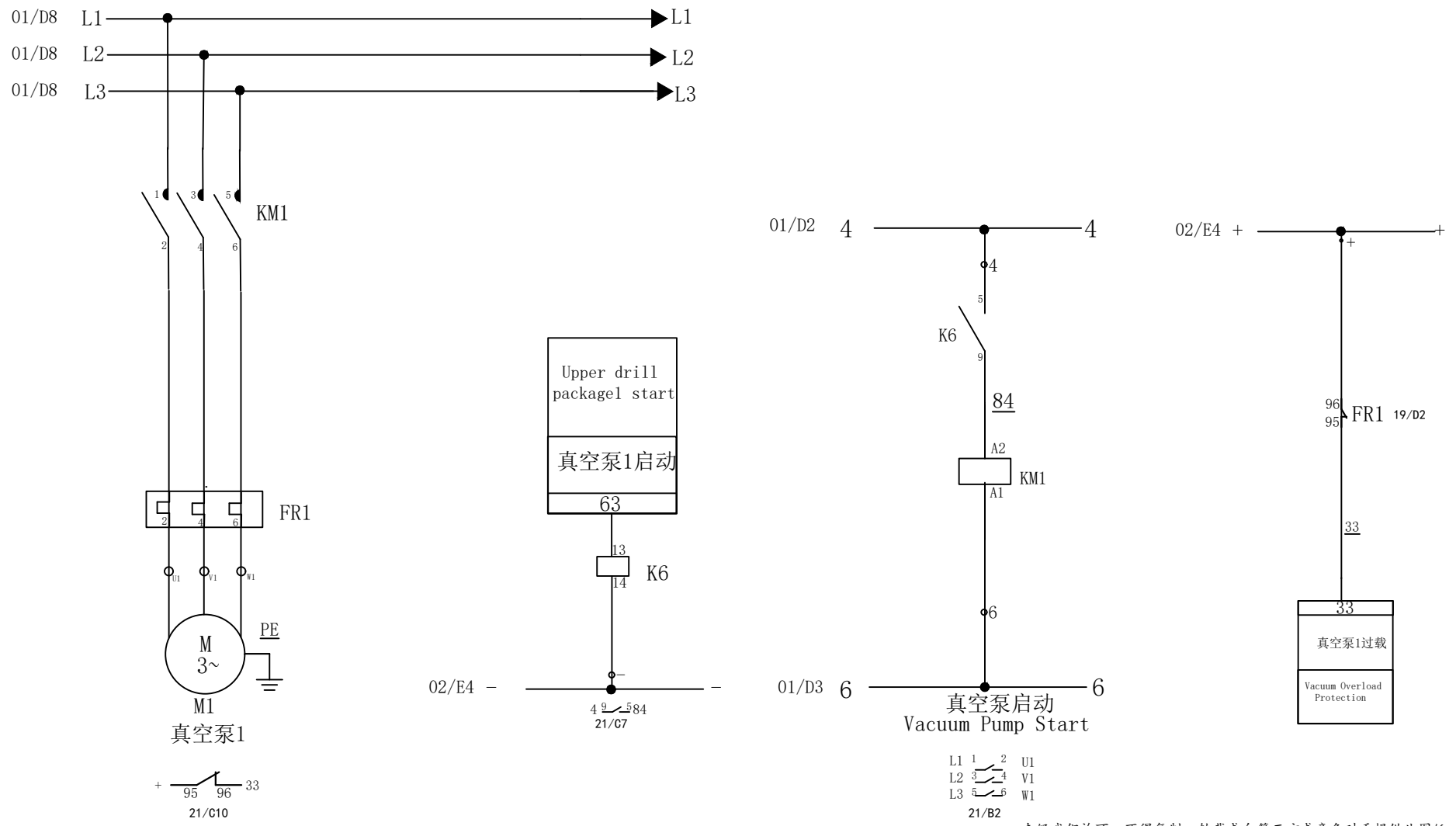
Electrical schematic diagram

除尘部分

Dust removal section

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HUAHUA®	电气原理图	真空泵1	
	Electrical schematic diagram	Vacuum Pump 1	页 Page: 21/21